

Assignment 1 – Computer Forensics

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Q1. Data Acquisition

a. Disk Image Creation

Step 1:

Select the “Create Disk Image...” within the “File” tab to start the process.

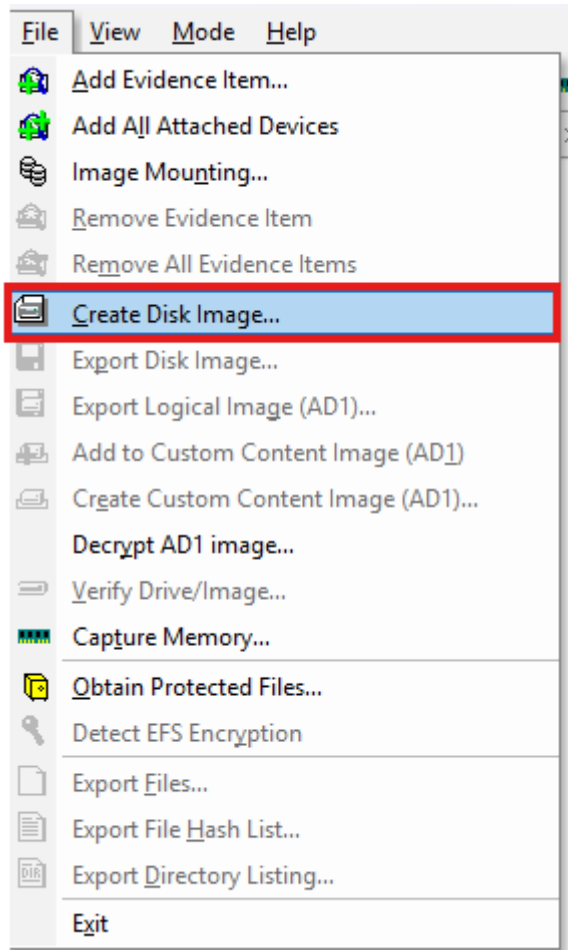


Figure 1: Creating new Disk Image

Step 2:

Select the “Physical Drive” option since we are creating a disk image from a physical USB drive.

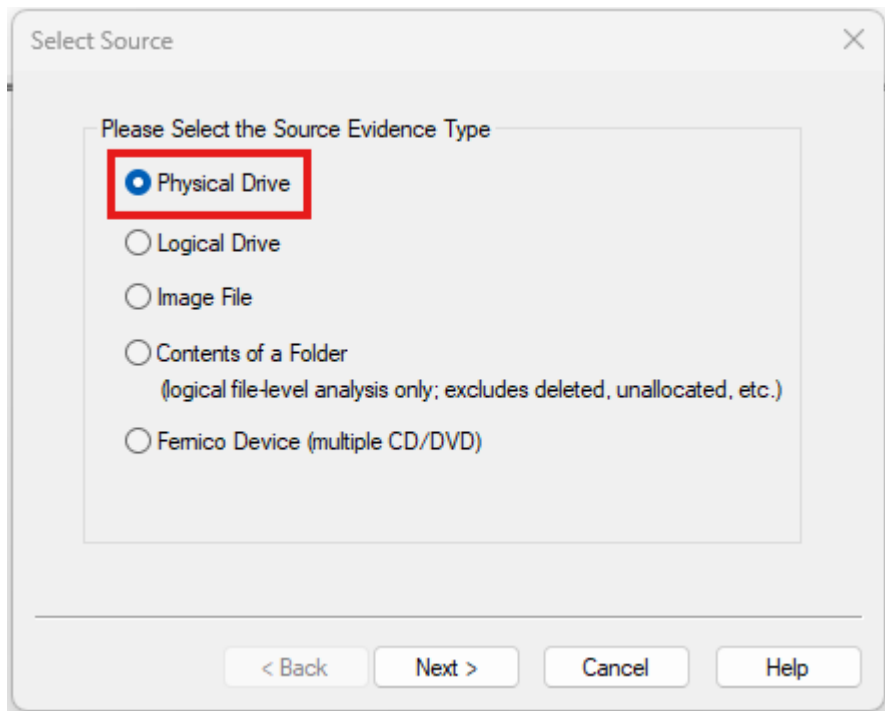


Figure 2: Selecting source

Step 3:

Selecting the appropriate drive for image creation.

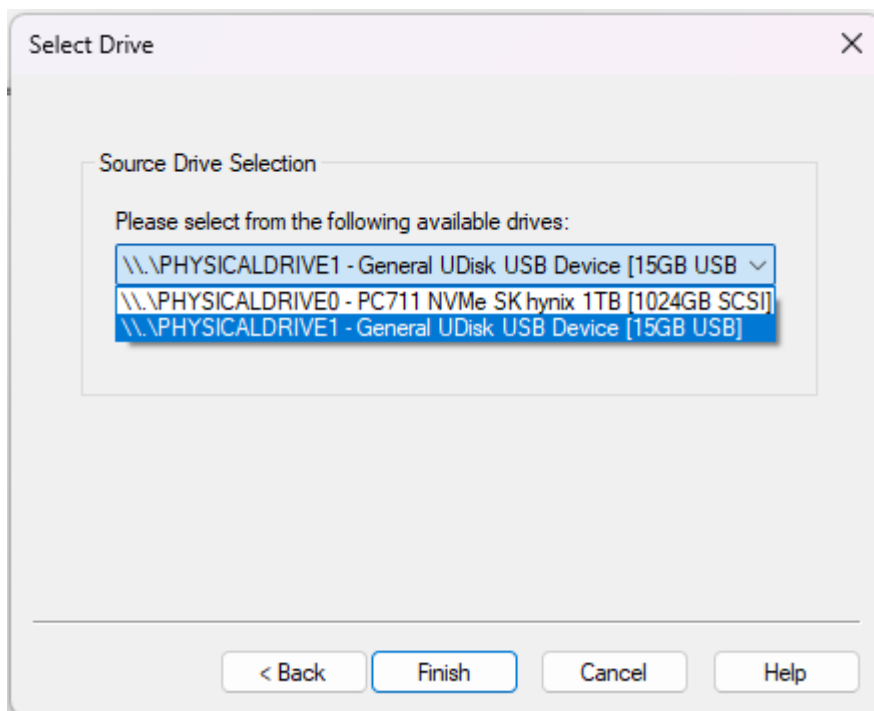


Figure 3: Selecting drive

Step 4:

Click add to configure the details of the drive image.

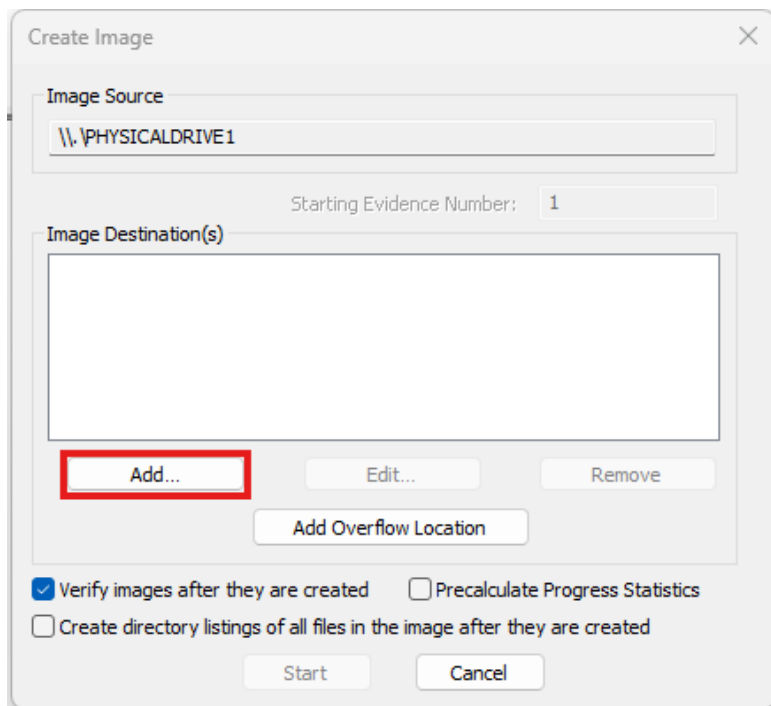


Figure 4: Configuring Image options

Step 5:

Select the type of the output image. We are selecting the “Raw (dd)” option, as requested within the assignment.

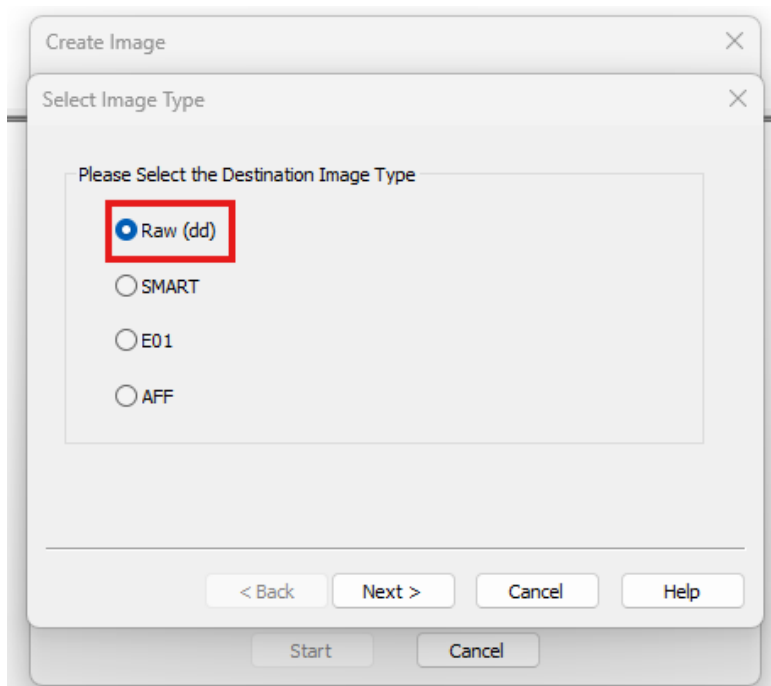
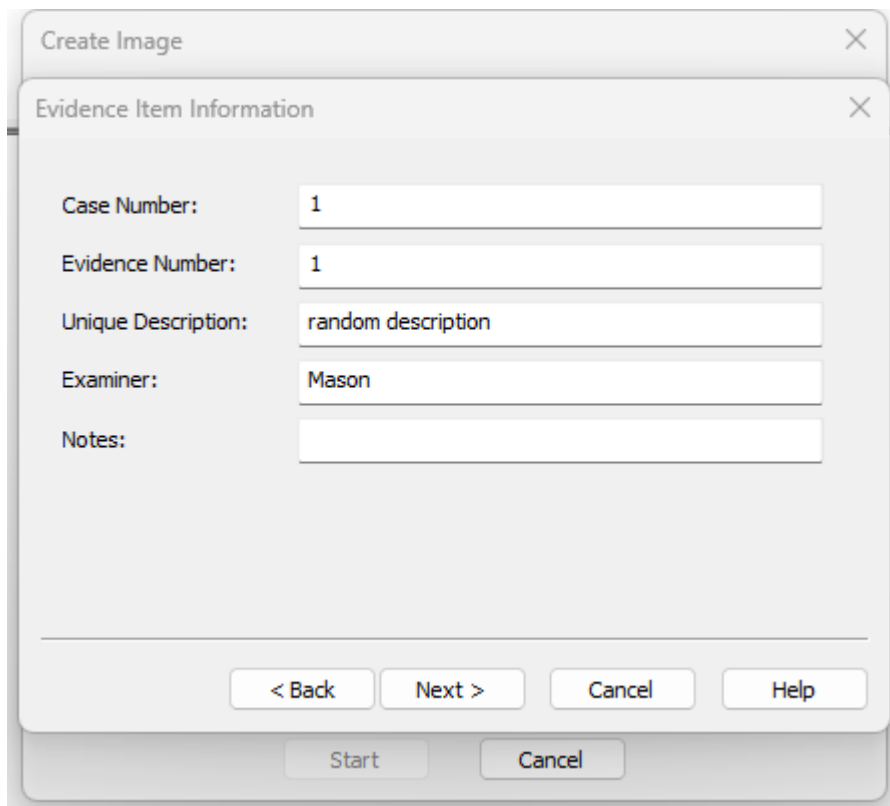


Figure 5: Selecting output Image type

Step 6:

Enter the details of the output image. No specific input is required in this step since the assignment didn't request any specific details about this part.



The screenshot shows a 'Create Image' dialog box with a sub-tab titled 'Evidence Item Information'. The sub-tab contains the following fields and values:

Field	Value
Case Number:	1
Evidence Number:	1
Unique Description:	random description
Examiner:	Mason
Notes:	

At the bottom of the 'Evidence Item Information' sub-tab, there are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'. Below the sub-tab, there are two more buttons: 'Start' and 'Cancel'.

Figure 6: Configuring details of Image

Step 7:

For this step, put the image filename as requested within the assignment specifications. Additionally, change the fragment size from a default of 1500 to 0 (also requested by the assignment). Also, I've put the destination folder as my desktop, but this is only due to my preference and makes no impact on the result.

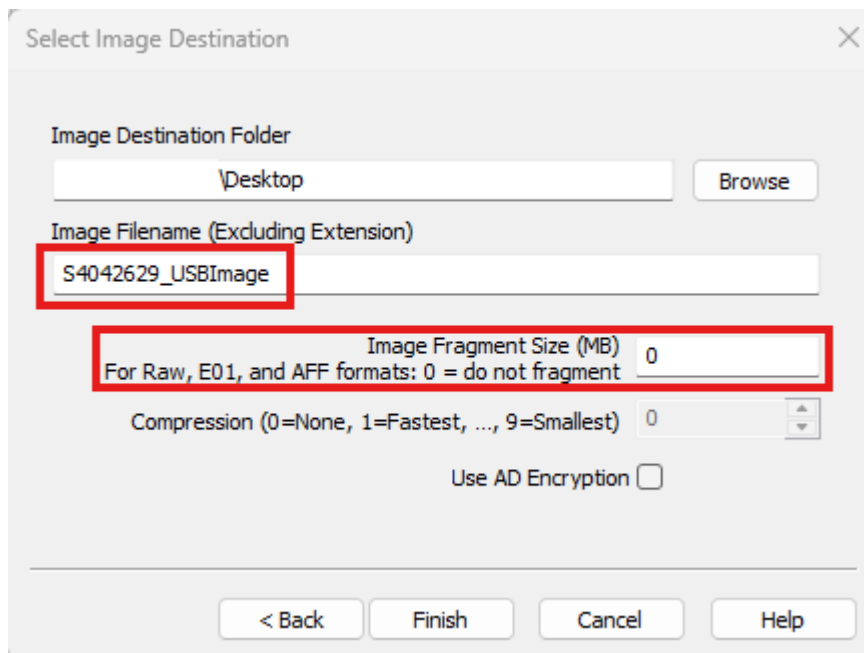


Figure 7: Configuring Image name, destination, and fragmentation

Step 8:

Recheck all the configuration of the disk image. Make sure “verify images after they are created” is ticked. Click “start” to start the process.

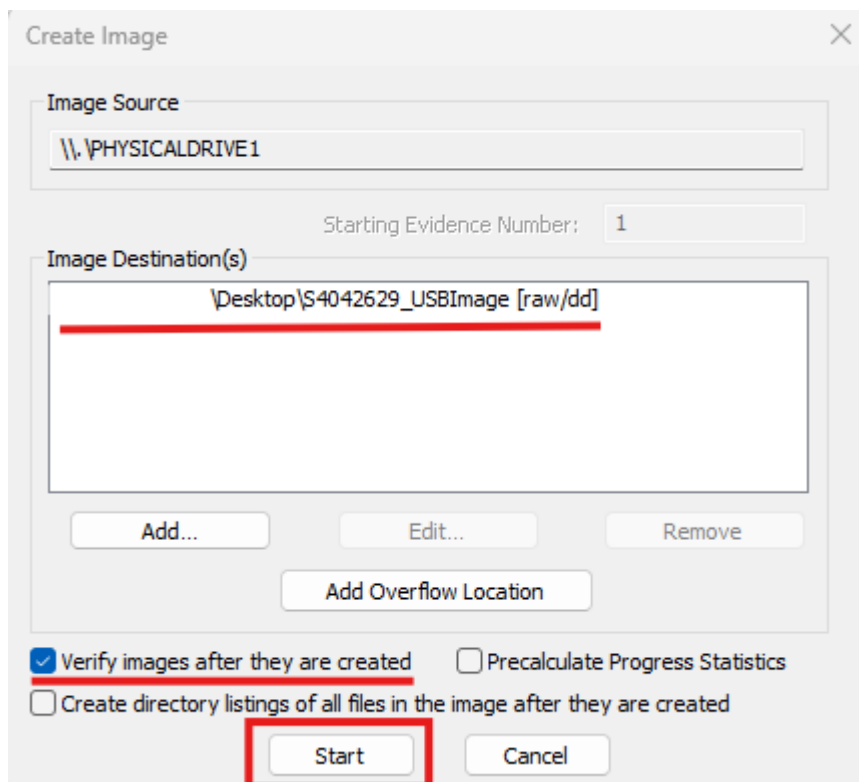


Figure 8: Starting Image creation

b. Disk Image information

The figure below showcases the information of the disk image, gathered from the .txt file.

```
Created By AccessData® FTK® Imager 4.7.1.2

Case Information:
Acquired using: ADI4.7.1.2
Case Number: 1
Evidence Number: 1
Unique description: random description
Examiner: Mason
Notes:

-----

Information for C:\[REDACTED]\Desktop\S4042629_USBImage:

Physical Evidentiary Item (Source) Information:
[Device Info]
  Source Type: Physical
[Drive Geometry]
  Cylinders: 1,912
  Tracks per Cylinder: 255
  Sectors per Track: 63
  Bytes per Sector: 512
  Sector Count: 30,720,000
[Physical Drive Information]
  Drive Model: General UDisk USB Device
  Drive Serial Number: 0901090447033862834317
  Drive Interface Type: USB
  Removable drive: True
  Source data size: 15000 MB
  Sector count: 30720000
[Computed Hashes]
  MD5 checksum: 5ca6503e059bd43306fd686b9fc33ce8
  SHA1 checksum: f8a97c570222ff108163a33afa2fbcaf12b9e7cb

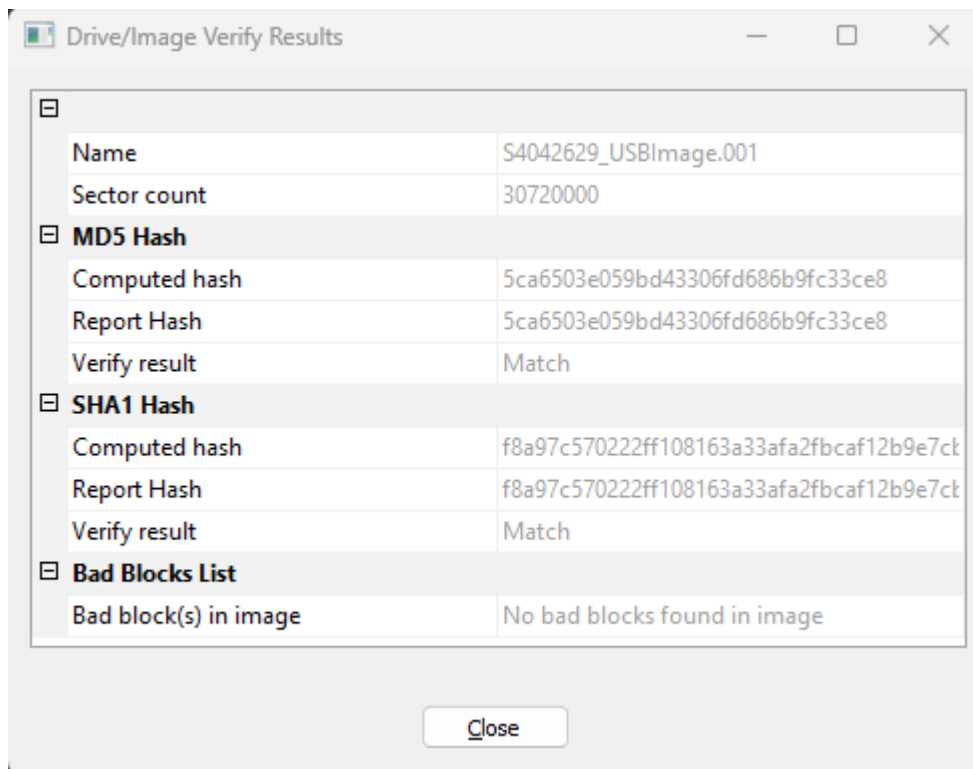
Image Information:
Acquisition started: Sat Aug 9 14:21:19 2025
Acquisition finished: Sat Aug 9 14:29:17 2025
Segment list:
  C:\[REDACTED]\Desktop\S4042629_USBImage.001

Image Verification Results:
Verification started: Sat Aug 9 14:29:18 2025
Verification finished: Sat Aug 9 14:29:58 2025
MD5 checksum: 5ca6503e059bd43306fd686b9fc33ce8 : verified
SHA1 checksum: f8a97c570222ff108163a33afa2fbcaf12b9e7cb : verified
```

Figure 9: Disk Image information

c. Questions

- i. The creation process started on Aug 9 14:21:19 2025 and finished on Aug 9 14:29:17 2025. The process took roughly 8 minutes to complete.



ii.

Figure 10: Image verification result

The checksums of the physical disk are:

- MD5 checksum: 5ca6503e059bd43306fd686b9fc33ce8
- SHA1 checksum: f8a97c570222ff108163a33afa2fbcaf12b9e7cb

iii.

The checksums of the image are:

- MD5 checksum: 5ca6503e059bd43306fd686b9fc33ce8 : verified
- SHA1 checksum: f8a97c570222ff108163a33afa2fbcaf12b9e7cb : verified

The verification of both checksums was successful, as evidenced by the word “verified” next to each checksum of the image, along with the verify result for each checksum showing a match, as presented in Figure 10.

Q2. Disk Image Analysis

a. Case Creation

Step 1:

Select the first option named “New Case” to start a new case.

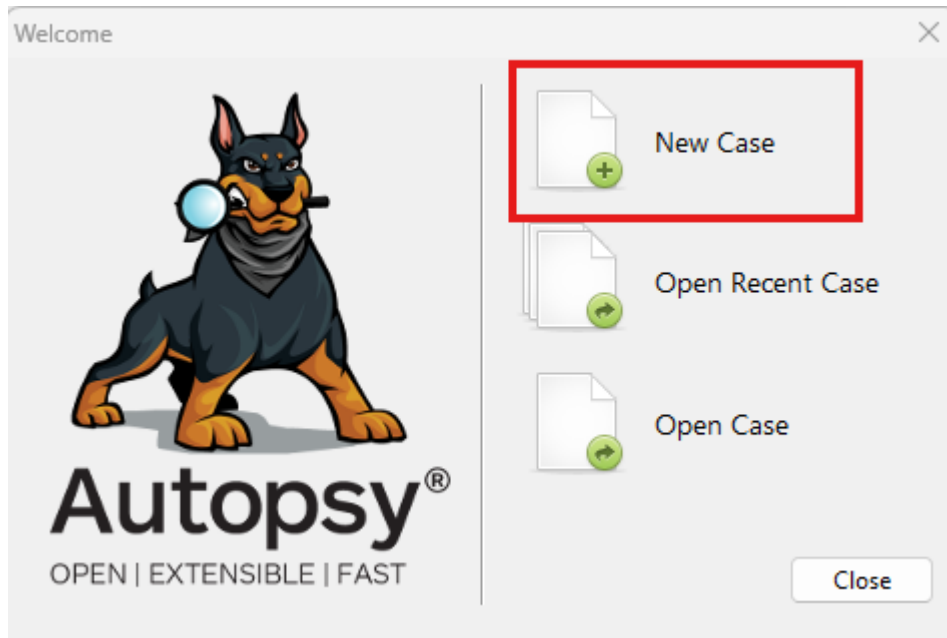
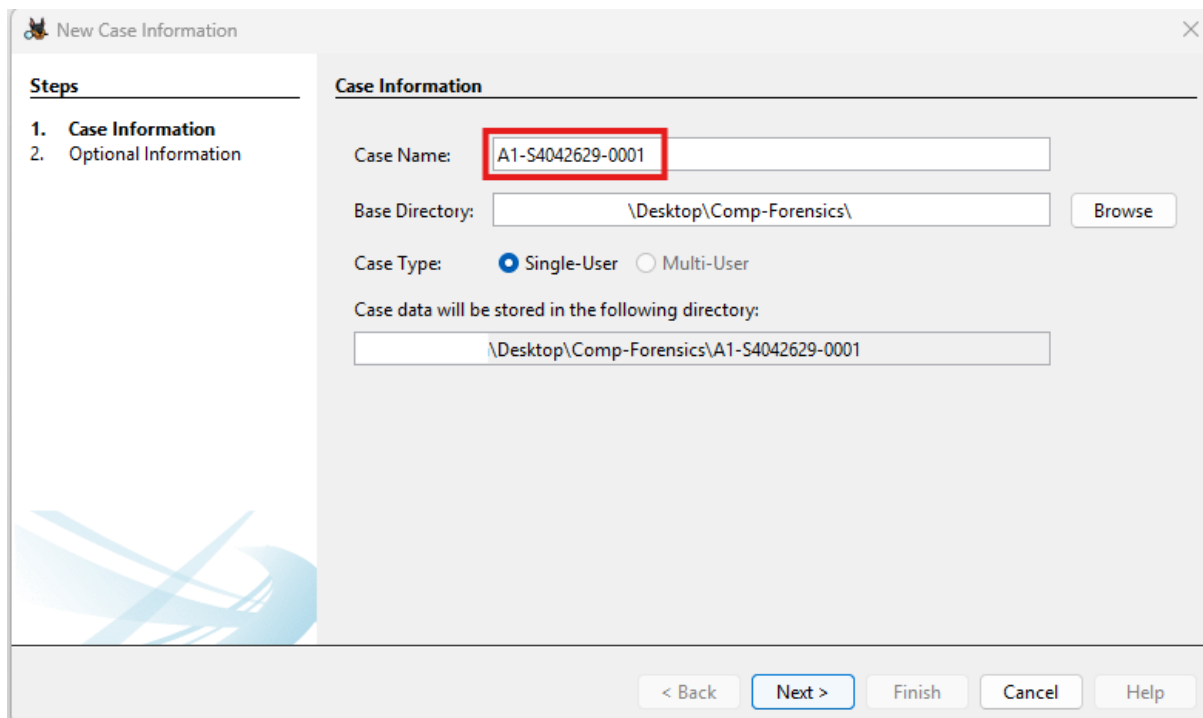


Figure 11: Creating new case

Step 2:

Enter the case name following the format described in the assignment details.

Additionally, configure the base directory (where the case contents will be saved) if this is your first time creating a new case.



New Case Information

Steps

1. **Case Information**
2. Optional Information

Case Information

Case Name:

Base Directory:

Case Type: ☒ Single-User ☐ Multi-User

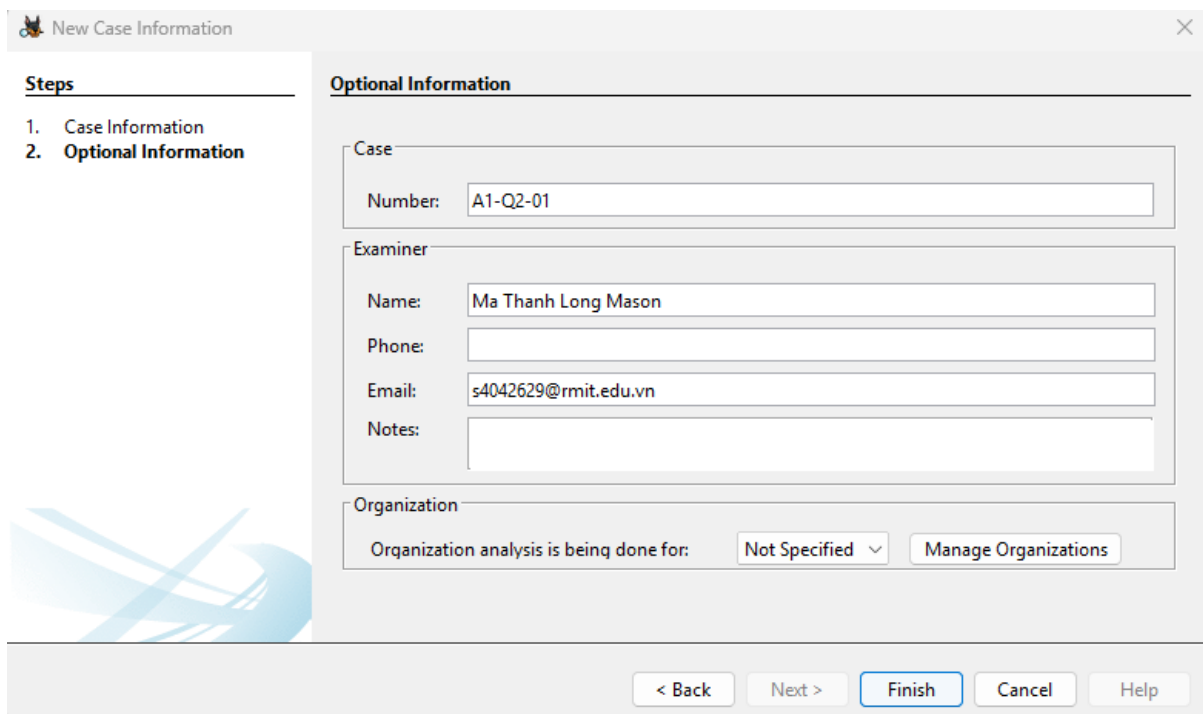
Case data will be stored in the following directory:

< Back **Next >** Finish Cancel Help

Figure 12: Configuring case name

Step 3:

Enter the requested optional information as shown in the figure above, then click “Finish” to proceed with the process.



New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number:

Examiner

Name:

Phone:

Email:

Notes:

Organization

Organization analysis is being done for:

< Back Next > **Finish** Cancel Help

Figure 13: Configuring optional information

Step 4:

Select the second bullet point and enter in host name in the requested format.

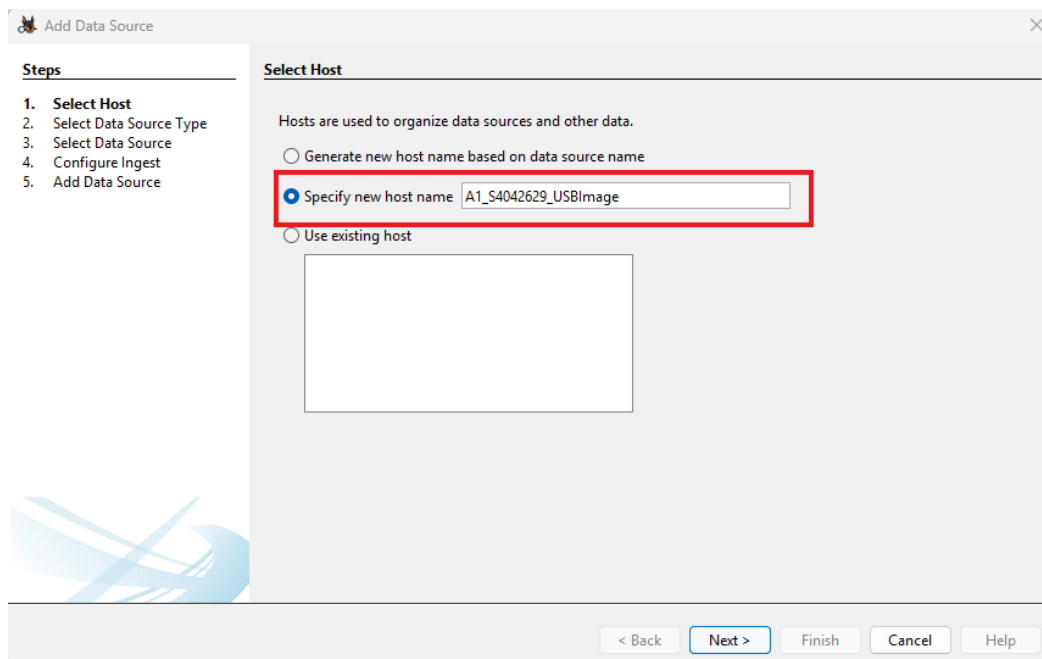


Figure 14: Configuring host name

Step 5:

Since we are trying to open and analyse a disk image (the one we created during Q1), we'll select the option "Disk Image or VM File" and click next to proceed.

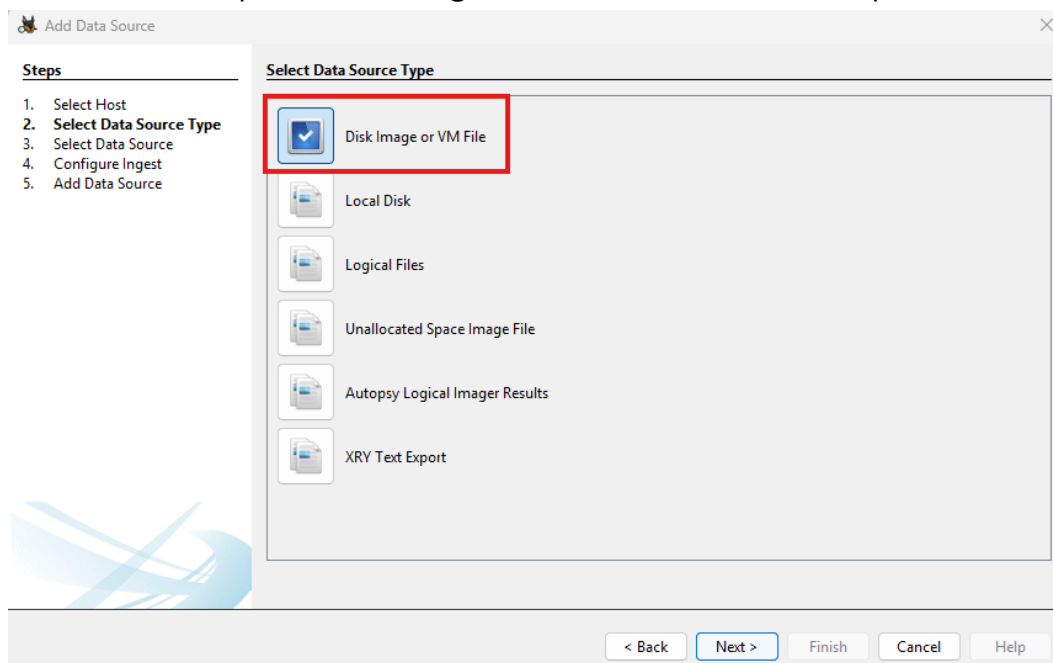


Figure 15: Selecting data source type

Step 6:

In this step, click on “Browse” and select the disk image created during Q1. Additionally, input the checksums into their appropriate text fields. Once finished, you can continue clicking on “Next” to proceed; leave everything in the next two pages as default and click on “Finish” once everything is done to conclude the case configuration.

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. **Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Path:

☐ Ignore orphan files in FAT file systems

Time zone:

Sector size:

Bitlocker Password (optional):

Hash Values (optional):

MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back Next > Finish Cancel Help

Figure 16: Entering information of data source

b. List of files

Table 1: List of files

File type	Amount
Images	3

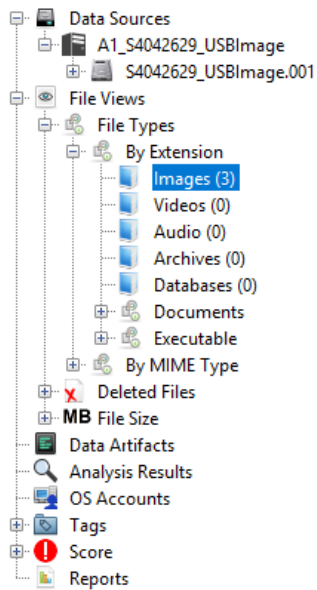


Figure 17: List of files

c. File extraction

Step 1:

Navigate the program by click on the various tabs, sections and locate the file you want to extract.



Figure 18: Navigating the Autopsy menu

Step 2:

Right click on the file you want to extract and select the “Extract File(s)” option to start.

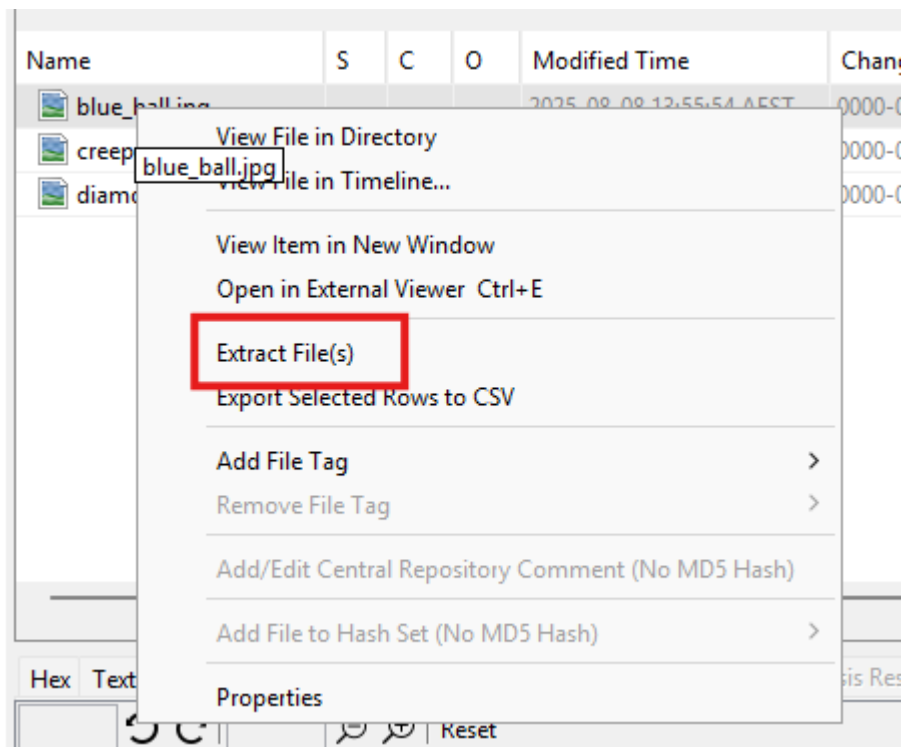


Figure 19: Extracting file

Step 3:

Once you click on the “Extract File(s)” on the previous step, a window will appear for you to select the location of the extracted file. You can change the export location or leave it as default, which will save the extracted file to a sub-folder named “Export”, located within the base directory of the case.

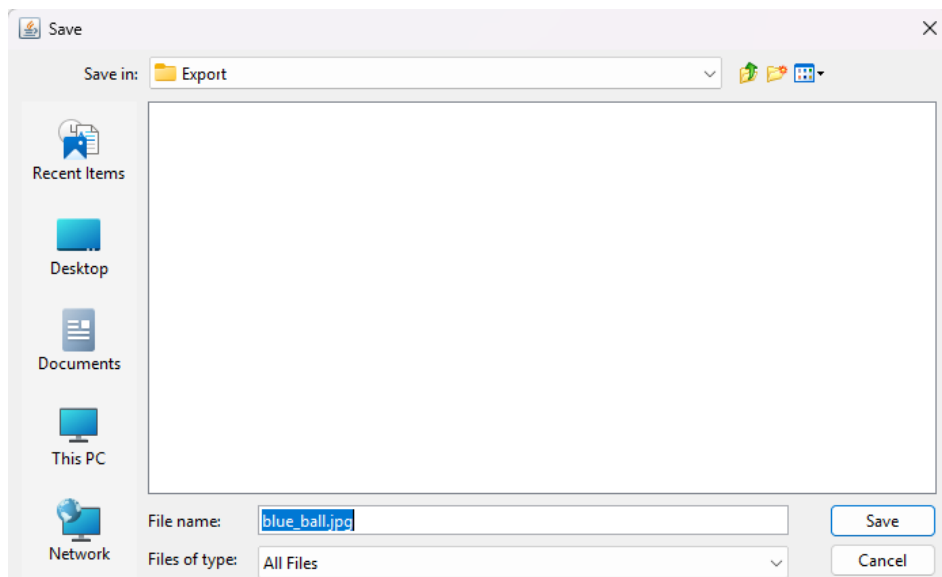


Figure 20: Saving extracted file

Q3. Understanding of Different Digital Forensics Tools

a. Static data acquisition tools

Tool	dc3dd	Guymager	OSFClone
OS	Linux & Windows	Linux	Linux & Windows
Pros	- Automatically detects bad sectors - Built-in hashing of input & output bytes - Integrates logging functionalities	- Can produce image files in raw, EWF, and AFF formats. - Multi-threaded - GUI-based program	- Bootable environment, independent of the installed operating system - Able to export in many formats such as: AFF, and EWF.
Cons	- CLI only, hard to use for beginners	- Linux only - Limited number of features beyond imaging	- May not be forensically sound when imaging drives with ext2/3/4 filesystems - No integrated analysis tools.

Table 2: Static data acquisition tools

b. Memory acquisition tools

Tool	Dumplt	LiME	WinPmem
OS	Windows	Linux & Andriod	Linux & Windows
Pros	- Portable, full physical memory capture - Minimal user interaction needed	- Acquires memory to file or over network - Supports ARM which is useful for mobile forensics - Open-source	- Supports multiple format - Integrated with Rekall for direct analysis - Open-source
Cons	- Windows-only - Always captures full RAM, which may not always be desired	- Requires loading a kernel module - Difficult to use	- Mostly CLI-based, not beginner-friendly - Has MacOS support but severely limited

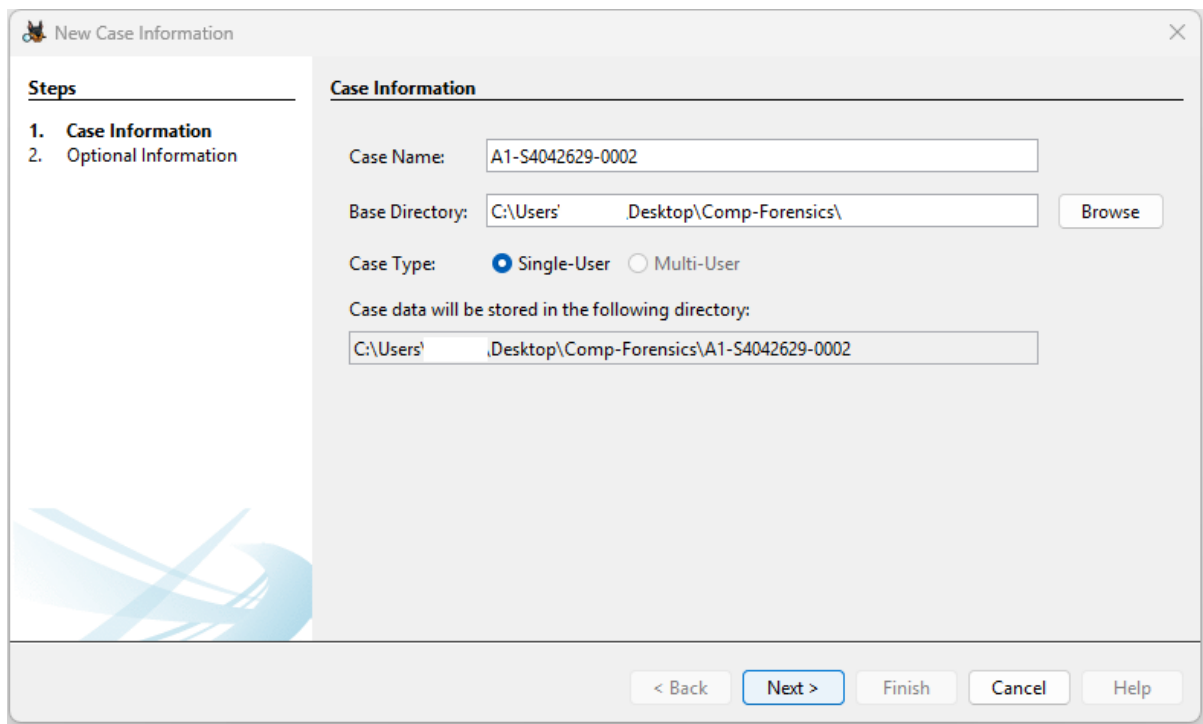
Table 3: Memory acquisition tools

Q4. Disk Image Artifact Analysis

a. File Listing and carving

Step 1:

Create a new case to start analysing the disk image. You can follow Figure 21 and Figure 22 to see what to put into the various input fields, but these are purely optional and doesn't impact the disk image analysis process.



New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name:

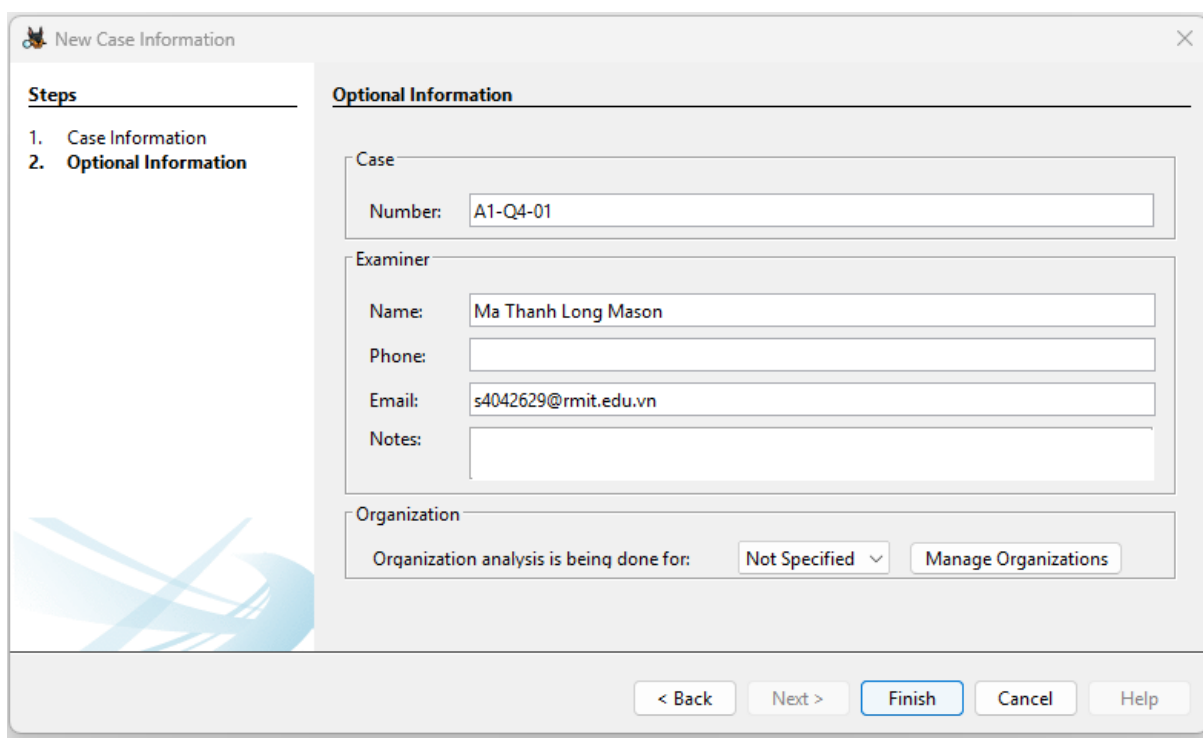
Base Directory:

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

< Back **Next >** Finish Cancel Help

Figure 21: Creating a new case



New Case Information

Steps

1. Case Information
2. Optional Information

Optional Information

Case

Number:

Examiner

Name:

Phone:

Email:

Notes:

Organization

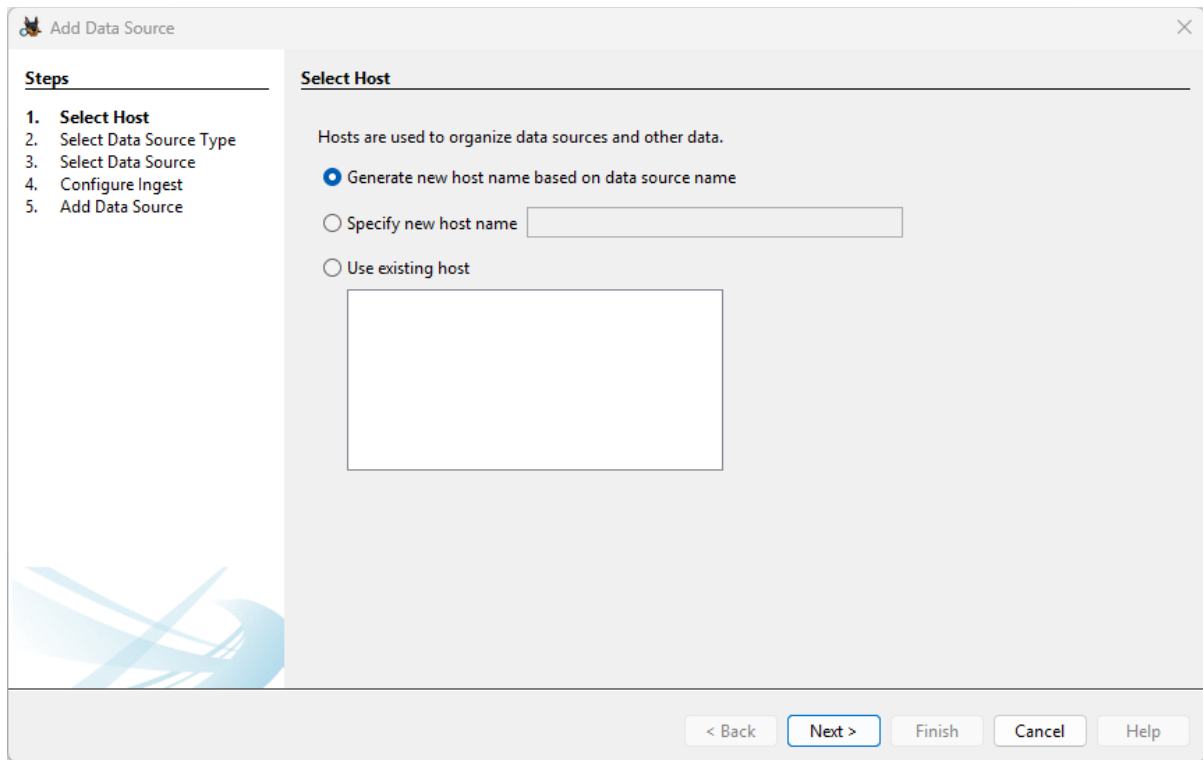
Organization analysis is being done for:

< Back Next > **Finish** Cancel Help

Figure 22: Entering optional information

Step 2:

Figures 23, 24, and 25 showcase the process of configuring the host and data source of the case. Follow the selected options in figures 23 and 24, which should be already selected by default, and for figure 25, click the browse button and select the disk image file.



The screenshot shows a window titled "Add Data Source" with a close button (X) in the top right corner. On the left, a "Steps" sidebar lists five steps: 1. Select Host (highlighted), 2. Select Data Source Type, 3. Select Data Source, 4. Configure Ingest, and 5. Add Data Source. The main area is titled "Select Host" and contains the text "Hosts are used to organize data sources and other data." Below this, there are three radio button options: "Generate new host name based on data source name" (which is selected), "Specify new host name" (with an adjacent text input field), and "Use existing host" (with an adjacent empty rectangular box). At the bottom right, there are five buttons: "< Back", "Next >" (highlighted with a blue border), "Finish", "Cancel", and "Help".

Figure 23: Selecting host

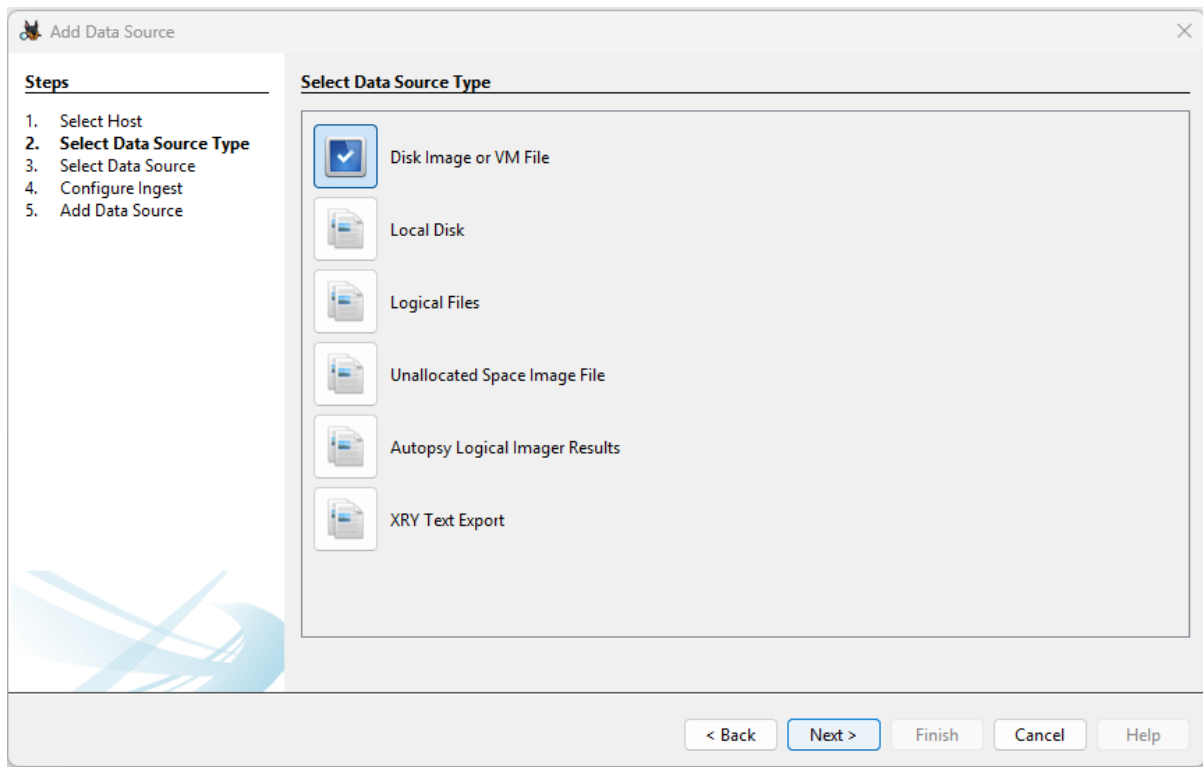


Figure 24: Selecting data source type

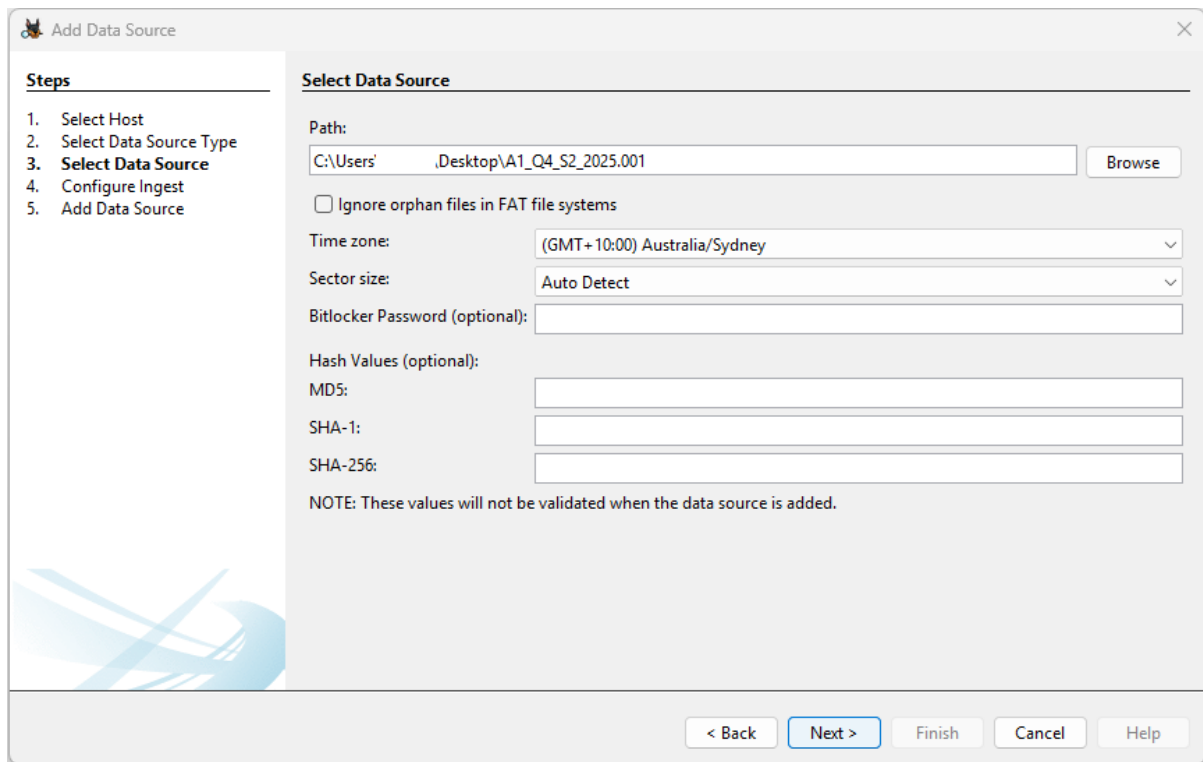


Figure 25: Selecting data source

Step 3:

Leave everything in this step as default, which should have all the modules selected except for “Cyber Triage Malware Scanner” and “Plaso”. Once you are done with this step, click “next” and on the next page click “Finish” to complete the case creation.

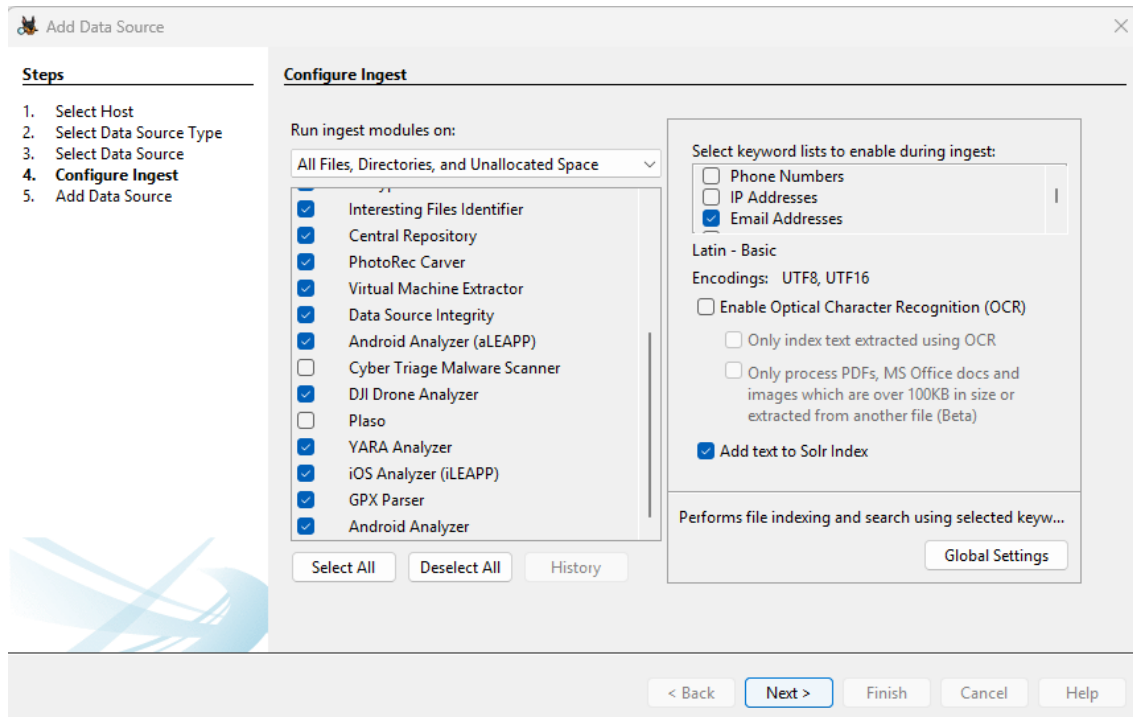


Figure 26: Configuring ingestion

Step 4

Once you have reached this step, you should now see the file contents of the disk image. However, as shown in figure 27, some tabs like the “By MIME Type” tab is missing its data, which hints at the ingestion being incomplete.

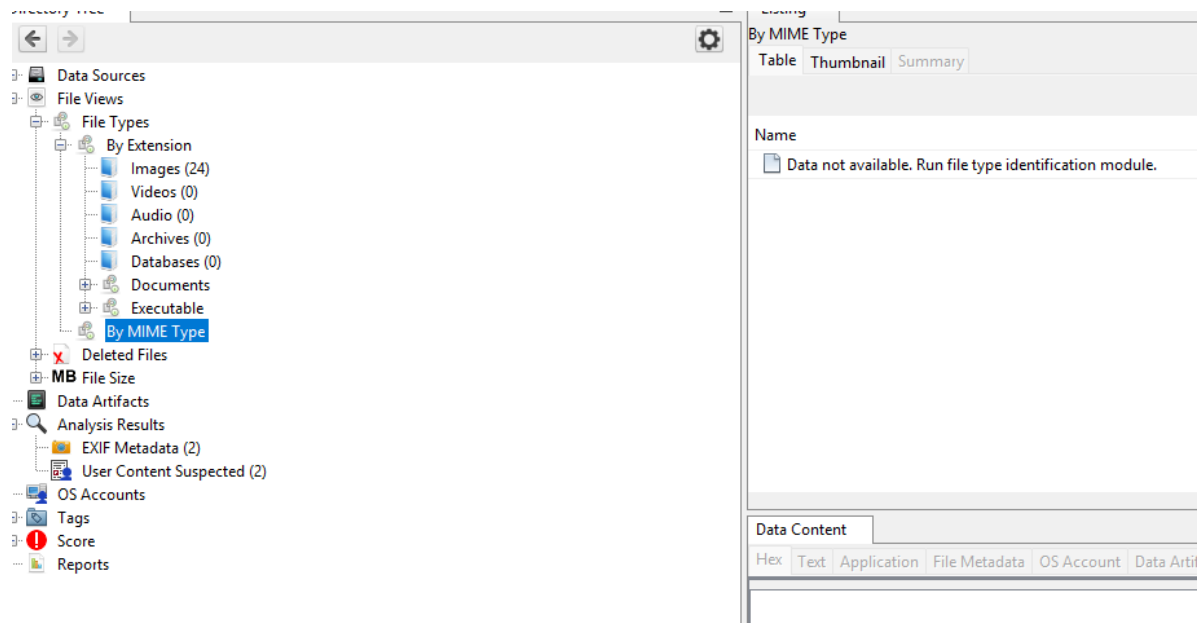


Figure 27: Disk image contents first look

Step 5:

We can try to fix this by navigating to the “tools” tab at the top of screen, navigating to “Run Ingest modules” and selecting the disk image file, as shown in figure 28.

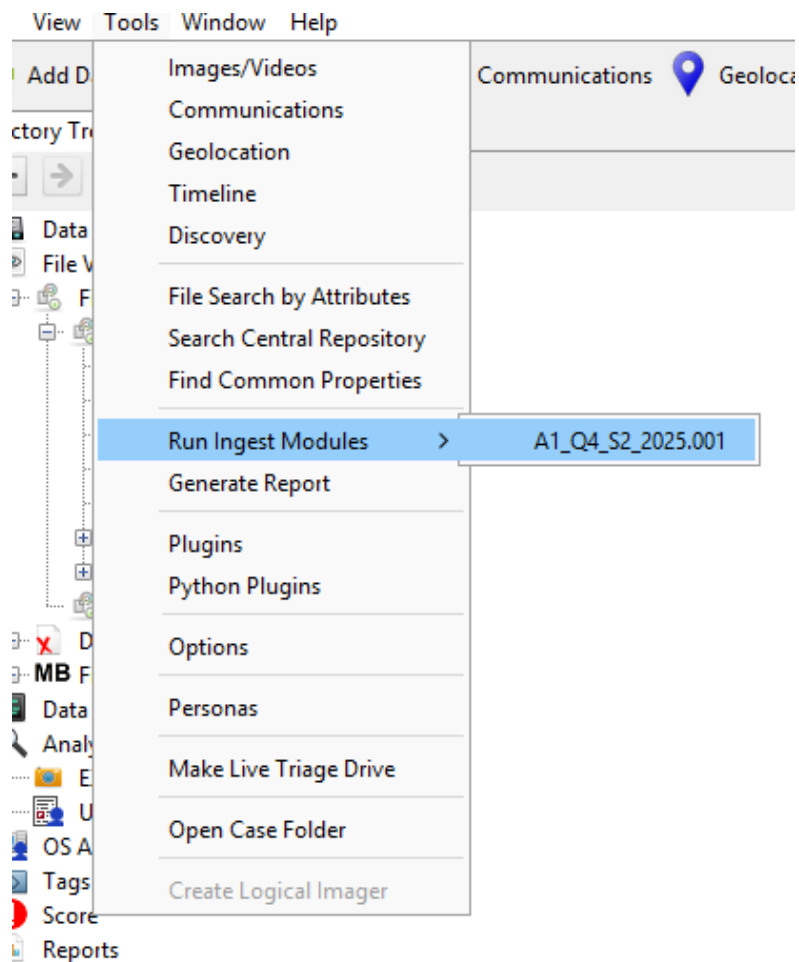


Figure 28: Running ingestion modules

Step 6:

Leave everything in the “Configure Ingest Modules” tab as default, as shown below, and click “Finish” to proceed with the ingestion.

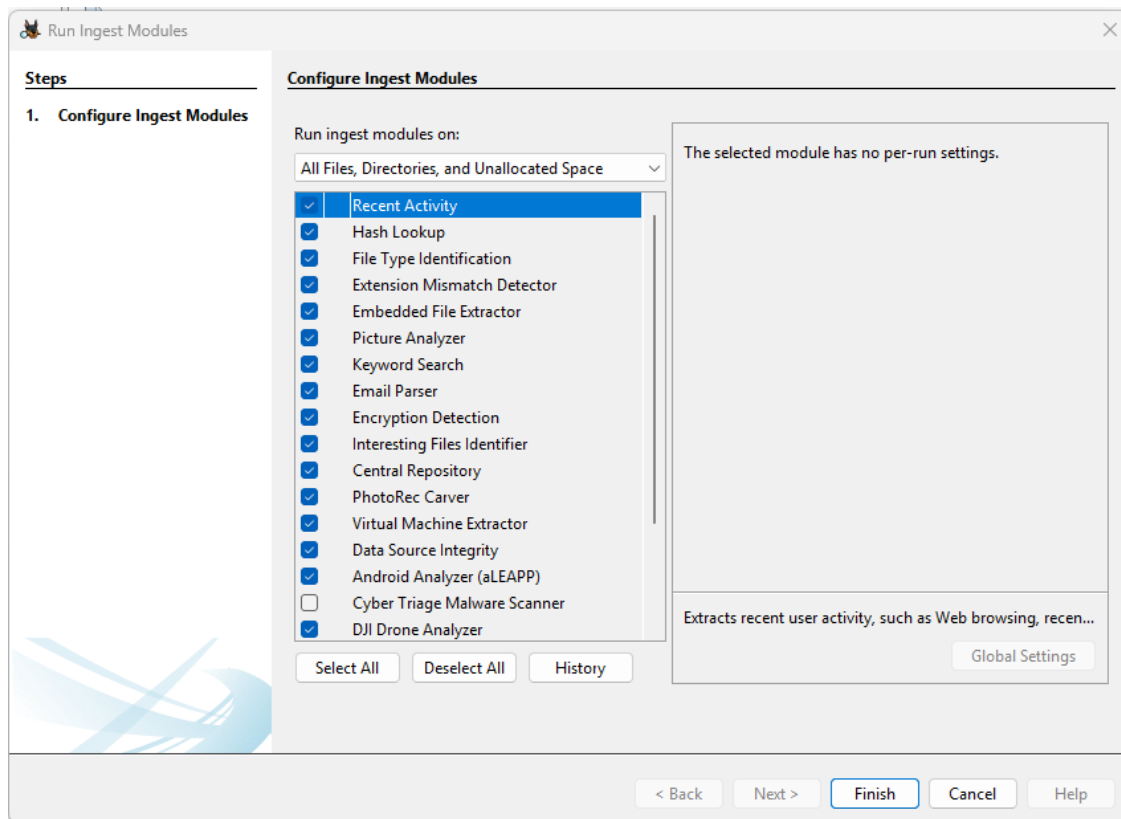


Figure 29: Configuring ingestion modules

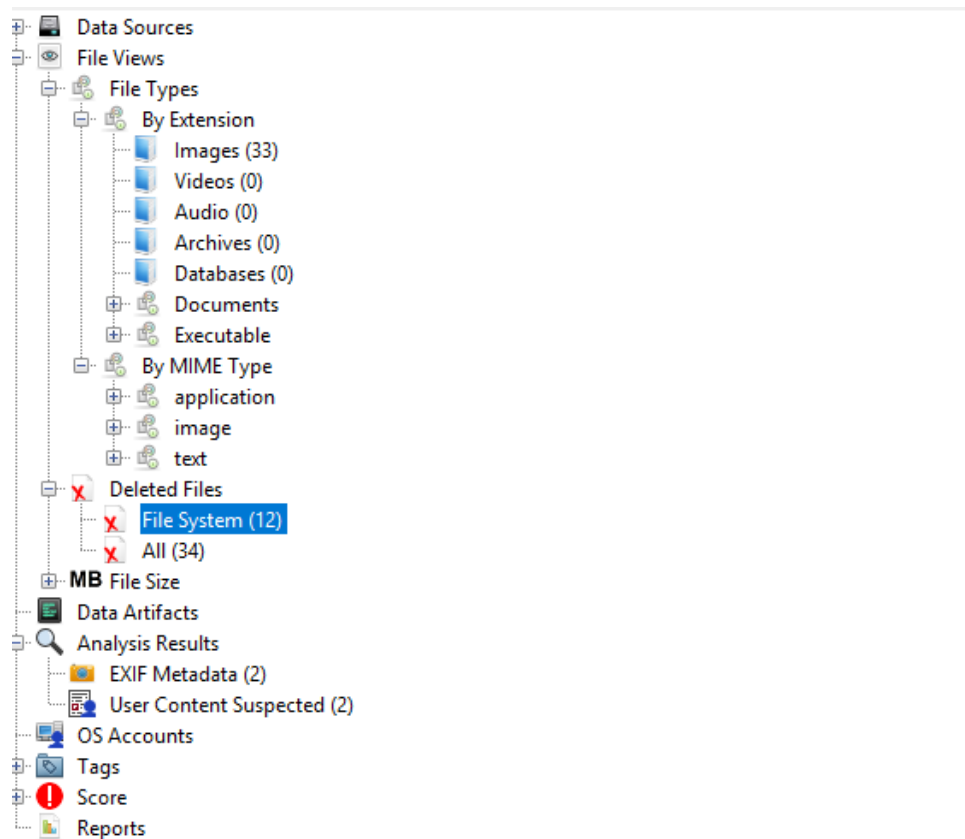


Figure 30: Complete ingestion of disk image

List of files by extension

Images

- animal5.jpg
- animal2.jpg
- animal6.jpg
- hashAlgo.jpg
- myfav.jpeg
- myfav.jpg
- f0000016.jpg
- f0000032.jpg
- f0000016.jpg
- f0000032.jpg
- animal1.png
- caesarcipher.png
- interesting1.png
- ref1.png
- rotCipher.png
- secretcode.png
- stag1.png
- stago1.png
- tools.png
- f0000056.png
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png
- f0000056.png
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png

Plain text

- AES.txt
- log1.txt

- log3.txt
- f0000392.txt
- f0000392.txt

List of files by MIME type

application/octet-stream

- \$AttrDef
- \$BadClus
- \$Bitmap
- \$Boot
- \$ObjId
- \$Quota
- \$Reparse
- \$Repair
- \$Repair:\$Config
- \$Tops:\$T
- \$TxfLog.blf
- \$TxfLogContainer0000000000000000000001
- \$TxfLogContainer0000000000000000000002
- \$LogFile
- \$MFT
- \$MFTMirr
- \$Volume
- Bankdetails.gpg
- IndexerVolumeGuid
- WPSettings.dat
- f0000000.DS_Store
- f0000000.DS_Store

image/vnd.microsoft.icon

- \$UpCase

image/vnd.zbrush.pcx

- \$Tops

List of deleted files

- .DS_Store
- AES (1).txt
- animal3.jpg
- animal4.jpg

- interesting.txt
- log2.txt
- ref2.png
- ref5.png
- saf1.png
- secretPlan.png
- stago_howitworks.png
- useful.png
- f0000000.DS_Store
- f0000016.jpg
- f0000032.jpg
- f0000056.png
- f0000392.txt
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png
- f0000000.DS_Store
- f0000016.jpg
- f0000032.jpg
- f0000056.png
- f0000392.txt
- f0000400.png
- f0002064.png
- f0002128.png
- f0006376.png
- f0008864.png
- f0009848.png

b. Finding Suspect's Hideout Location

Step 1:

Perform a keyword search to identify the suspicious files.

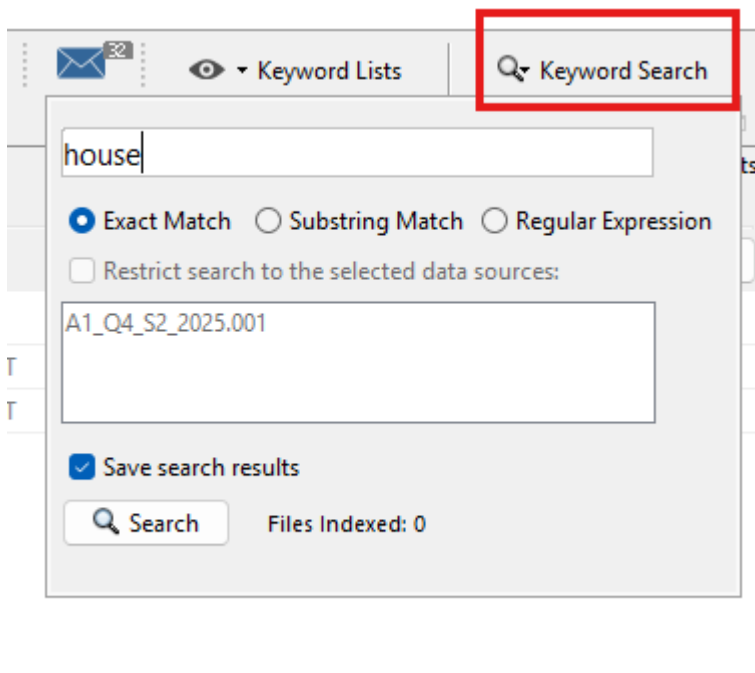


Figure 31: Keyword Search

Name	Keyword Preview	Location	M
myfav.jpeg	ExifcuteTaken near my «House»Photoshop 3.08BIMTa...	/img_A1_Q4_S2_2025.001/vol_vol2/myfav.jpeg	20:
myfav.jpg	ExifcuteTaken near my «House»!:%!&8&+ /1555\$;@;4?....	/img_A1_Q4_S2_2025.001/vol_vol2/myfav.jpg	20:

Figure 32: Suspicious files

Step 2:

Navigate to the suspicious files and select the “Analysis Results” tab. In said tab, the latitude and longitude of where the picture was taken should be displayed if available.

File Name	Size	Created	Modified	Accessed	Permissions	Location
myfav.jpg	1	2025-08-02 12:29:50 AEST	2025-08-02 12:32:09 AEST	2025-08-03 02:10:21 AEST	2025-08-03 02:10:21 AEST	9161 Allocated Allocated unknown /img_A1_Q4_S2_2025.001/vol_vol2/myfav.jpg
myfav.jpeg	1	2025-08-02 12:29:55 AEST	2025-08-02 12:32:09 AEST	2025-08-03 02:10:21 AEST	2025-08-03 02:10:21 AEST	19654 Allocated Allocated unknown /img_A1_Q4_S2_2025.001/vol_vol2/myfav.jpeg
interesting1.png	1	2025-08-02 12:29:50 AEST	2025-08-03 00:57:32 AEST	2025-08-03 02:10:21 AEST	2025-08-03 02:10:21 AEST	20787 Allocated Allocated unknown /img_A1_Q4_S2_2025.001/vol_vol2/interesting1.png
r0002064.png	1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	30608 Unallocated Unallocated unknown /img_A1_Q4_S2_2025.001/vol_vol2/r0002064.png
r0002064.png	2	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	30608 Unallocated Unallocated unknown /img_A1_Q4_S2_2025.001/vol_vol2/r0002064.png

Hex Text Application File Metadata OS Account Data Artifacts **Analysis Results** Context Annotations Other Occurrences

Item: myfav.jpg
Aggregate Score: Not Notable

Analysis Result 1
Score: Not Notable
Type: EXIF Metadata
Configuration:
Conclusion:
Justification:
Latitude: -33.80403870477934
Longitude: 149.77585577058966

Analysis Result 2
Score: Likely Notable
Type: Keyword Hits
Configuration:
Conclusion:
Justification:
Keywords: near
Keyword Preview: jpg JFIF Exif cute Taken «near» my House!:%! &8&+ /1555
Keyword Search Type: 0

Analysis Result 3
Score: Likely Notable
Type: Keyword Hits
Configuration:

Figure 33: Coordinates in image

Result

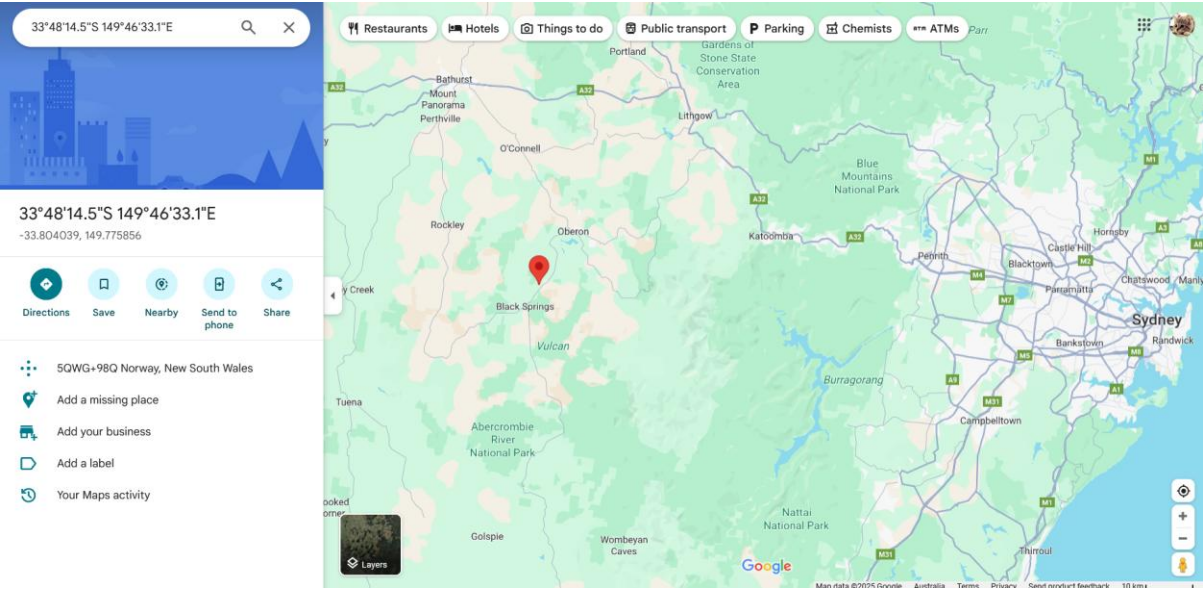


Figure 34: Extracted coordinates in google maps

c. Decryption of Bank Login Details

Step 1:

We first locate the suspicious file by browsing through all the different files on the disk image and isolating the most promising one.

Bankdetails.gpg	1	2025-08-02 12:29:53 AEST	2025-08-02 12:32:09 AEST	2025-08-03 02:10:21 AEST	2025-08-03 02:10:21 AEST	241	Allocated	Allocated	unknown	/img_A1
IndexerVolumeGuid	1	2025-08-03 02:08:18 AEST	2025-08-03 02:08:18 AEST	2025-08-03 02:08:18 AEST	2025-08-03 02:08:18 AEST	76	Allocated	Allocated	unknown	/img_A1

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Metadata

Name:/img_A1_Q4_S2_2025.001/vol_vol2/Bankdetails.gpg

TypeFile System

MIME Typeapplication/octet-stream

Size241

File Name AllocationAllocated

Metadata AllocationAllocated

Modified2025-08-02 12:29:53 AEST

Accessed2025-08-03 02:10:21 AEST

Created2025-08-03 02:10:21 AEST

Changed2025-08-02 12:32:09 AEST

MD524aa7383d7d4699b43ef08d50470bf3

SHA-256951b24905a65896042cde75950dc9c321690289585c09bb8378c427ddadce3d4

Hash Lookup ResultsUNKNOWN

Internal ID75

From The Sleuth Kit istat Tool:

MFT Entry Header Values:

Entry: 48

Sequence: 1

\$LogFile Sequence Number: 12589492

Allocated File

Links: 1

\$STANDARD_INFORMATION Attribute Values:

Flags: Archive

Figure 35: Potential bank details

Step 2:

Once we have located the suspicious file, we will browse through the disk image again to find clues on how to decrypt the file. The image shown in the figure below shows a command line that matches the tail “:gpg” of the suspicious file. By analysing the command, we can deduce that the file was possibly encrypted using symmetric encryption thus requires a passphrase to decrypt. Furthermore, we can see that AES256 was the algorithm used to encrypt the file.



Figure 36: Encryption command

Step 3:

A file named “AES”, located within the plain text section could potentially be related to the command shown in figure 36. We can try brute forcing the answer by putting the extracted text into different cipher-solving tools to decrypt the text. The text was encrypted using the ROT cipher as shown in the two figures below.

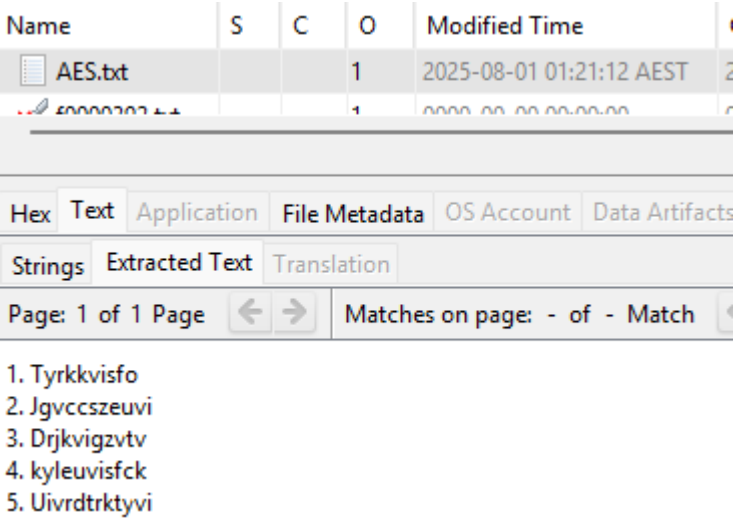


Figure 37: Encrypted passphrase



Figure 38: Decrypted passphrase

Step 4:

Another image file shows the command to decrypt the bankdetails file. We can proceed by going to KaliLinux and running the command, substituting the “sample.txt.gpg” with “Bankdetails.gpg” as shown in the two figures below. Once we run the command, the system will ask us for a passphrase to decrypt the file. We can now try to brute force the passphrase by trying all the 5 phrases we got from decrypting the “AES” text file.

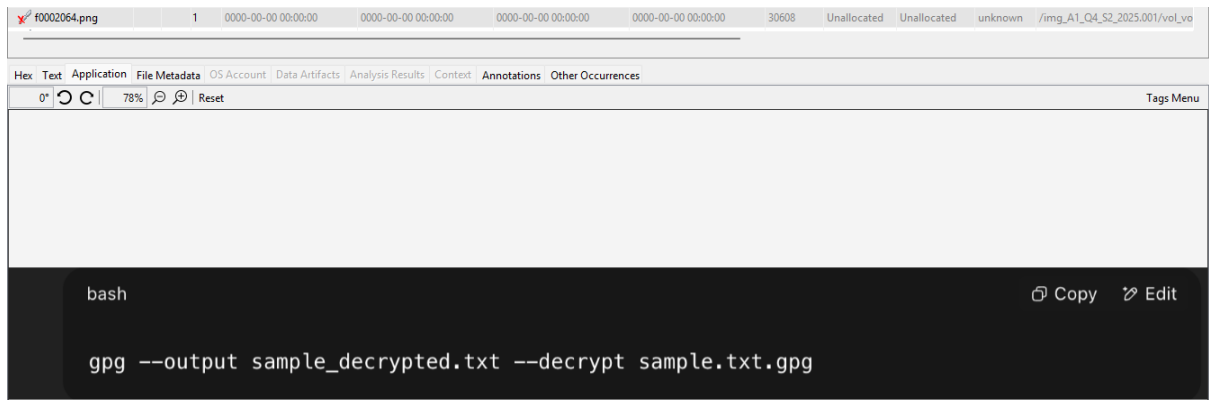


Figure 39: Gpg decrypt command

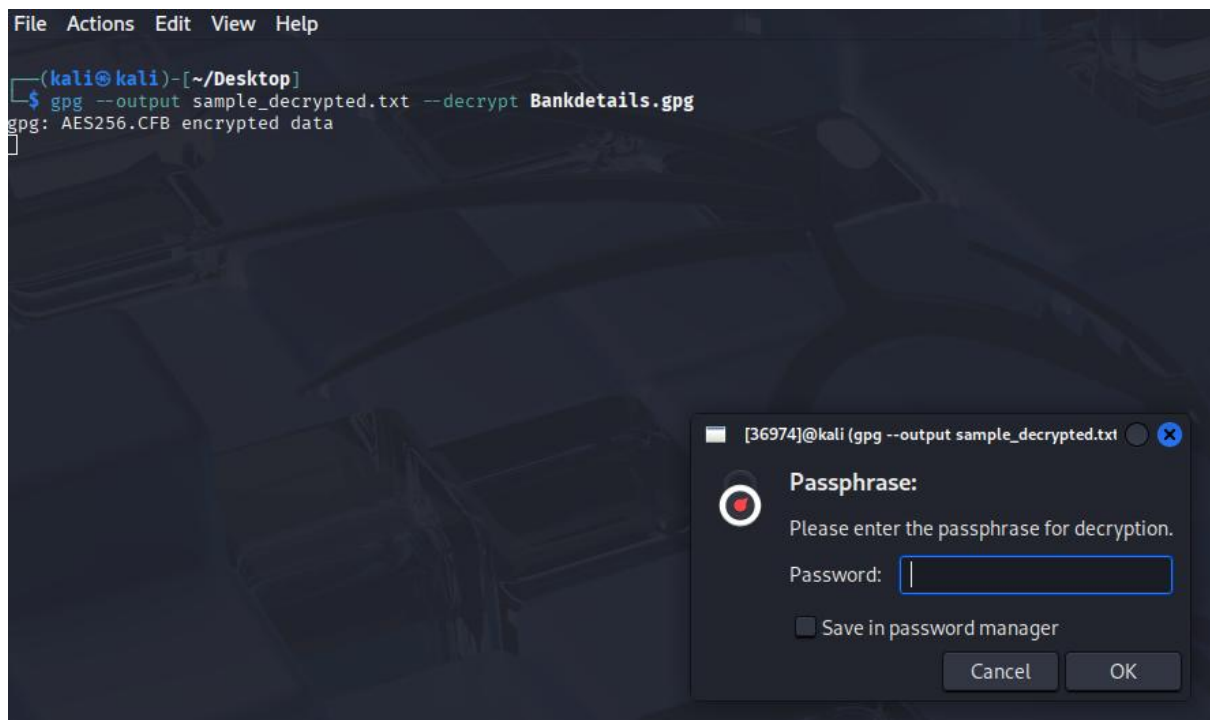


Figure 40: Prompted for passphrase

Result

The correct passphrase was “thunderbolt”. The contents of the file is shown in the figure below.

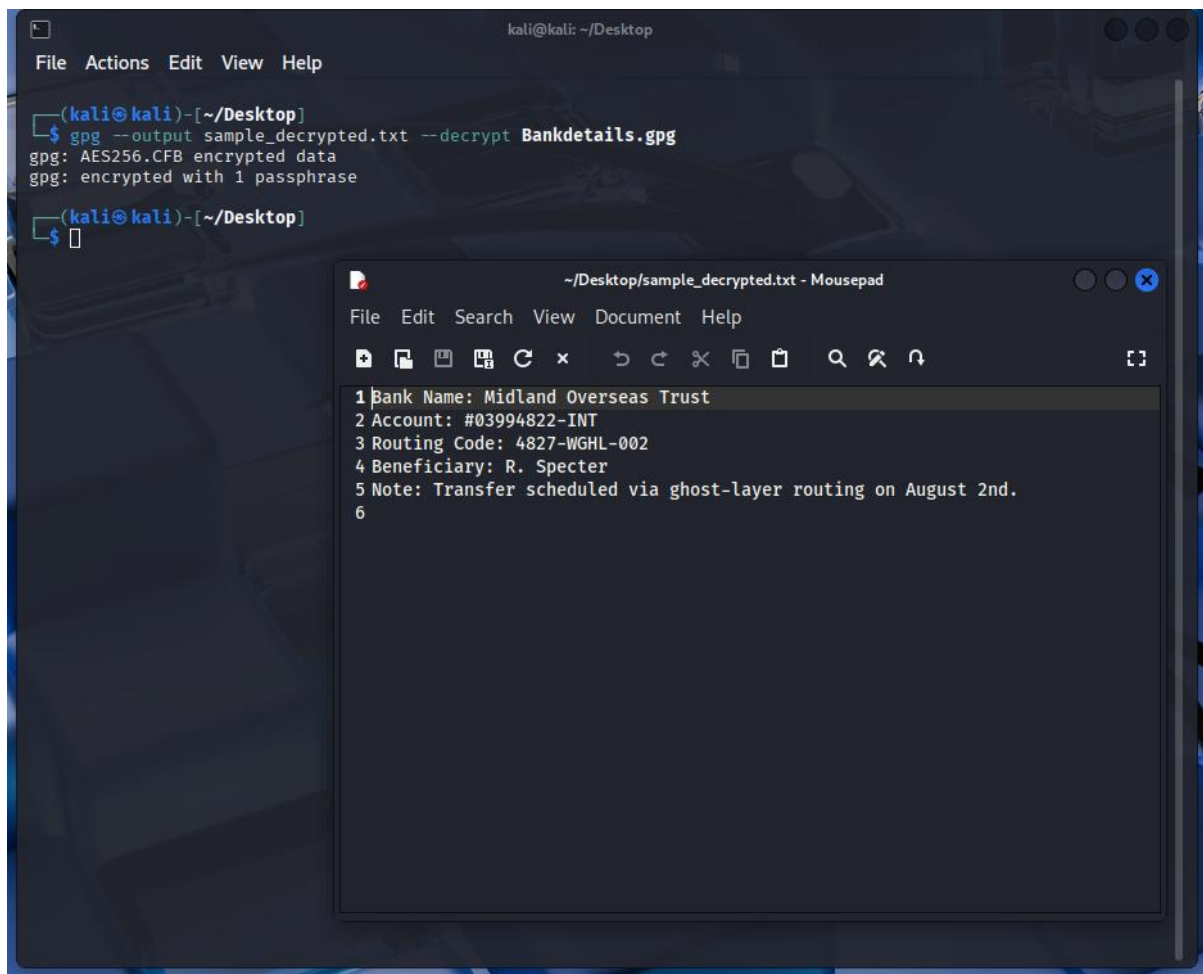


Figure 41: Decrypted bank details

d. Recovering the Vault PIN and Drawer Password

Step 1:

By browsing through all the files present in the disk image, we can find a suspicious .txt file which displays an image when we click on it.

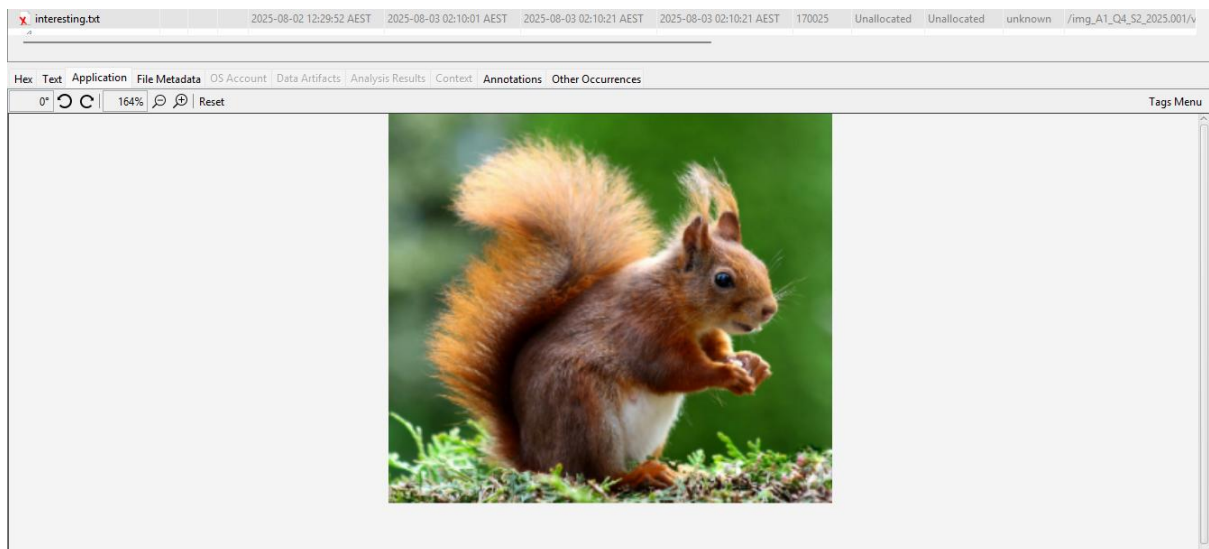


Figure 42: Suspicious file

If we examine the MIME type in metadata of the file, we can see that the file is in fact a .png file instead of a .txt file.

Metadata	
Name:	/img_A1_Q4_S2_2025.001/vol_vol2/interesting.txt
Type:	File System
MIME Type:	image/png
Size:	170025

Figure 43: Mismatching MIME Type

Step 2:

After exporting the suspicious file and running it through a steganography decrypting tool, we can see the hidden picture.



Figure 44: Decrypting image

Step 3:

We'll first decrypt the squares and triangles by using morse code. By mapping the hidden message onto the table provided in the figure below, we can make out the hidden message on the second and third line (excluding the login details line) to be "9910" and "75" respectively.

secretcode.png	1	2025-08-02 12:29:50 AEST	2025-08-03 01:00:39 AEST	2025-08-03 02:10:44 AEST	2025-08-03 02:10:22 AEST	73678	Allocated	Allocated	unknown
Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
0"	🔄	🔄	🔄	104%	🔍	🔍	Reset		
a	⬭	i	⬮	q	⬭⬭	y	⬭⬭	]	⬭⬭
b	⬭⬭⬭	j	⬭⬭⬭	r	⬭⬭	z	⬭⬭	;	⬭⬭
c	⬭⬭⬭	k	⬭⬭	s	⬭⬭	"	⬭⬭⬭	1	⬭⬭⬭
d	⬭⬭	l	⬭⬭⬭	t	⬭	(	⬭⬭⬭	2	⬭⬭⬭
e	⬭	m	⬭	u	⬭⬭	)	⬭⬭⬭⬭	3	⬭⬭⬭
f	⬭⬭⬭	n	⬭	v	⬭⬭⬭	{	⬭⬭⬭	4	⬭⬭⬭
g	⬭⬭	o	⬭	w	⬭⬭	}	⬭⬭⬭	5	⬭⬭⬭⬭⬭
h	⬭⬭⬭⬭	p	⬭⬭⬭	x	⬭⬭⬭	[	⬭⬭⬭	6	⬭⬭⬭⬭

Figure 45: Morse code table

Step 4:

We'll now decrypt the remaining message using the Caesar cipher to get the final decrypted message.

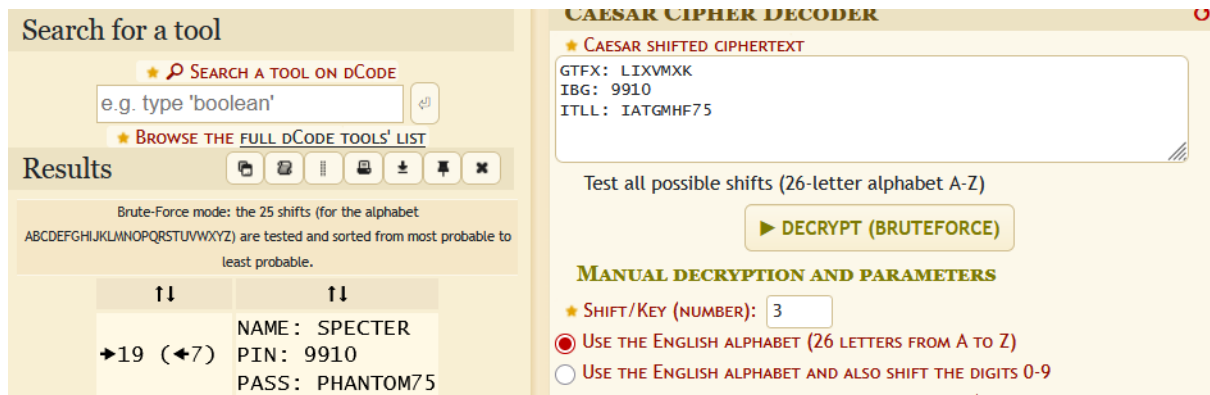


Figure 46: Decrypting hidden text

Result

NAME: SPECTER

PIN: 9910

PASSWORD: PHANTOM75

Q5. Memory Data Analysis from Computer Memory

a. Memory dump's basic information.

The figures below showcase the command to acquire the memory dump's basic information along with its output.

```
(kali㉿kali)-[~/volatility3]  
$ python3 vol.py -f a1memorydump.mem windows.info
```

Figure 47: Command to show memory dump's basic information

Variable	Value
Kernel Base	0x8183a000
DTB	0x122000
Symbols file	file:///home/kali/volatility3/volatility3/symbols/windows/ntkrpamp.pdb/37D328E3BAE5460F8E662756ED80951D-2.json.xz
Is64Bit	False
IsPAE	True
layer_name	0 WindowsIntelPAE
memory_layer	1 FileLayer
KdDebuggerDataBlock	0x81931c90
NTBuildLab	6001.18000.x86fre.longhorn_rtm.0
CSDVersion	1
KdVersionBlock	0x81931c68
Major/Minor	15.6001
MachineType	332
KeNumberProcessors	3405774849
SystemTime	2014-01-08 17:54:20+00:00
NtSystemRoot	C:\Windows
NtProductType	NtProductServer
NtMajorVersion	6
NtMinorVersion	0
PE MajorOperatingSystemVersion	6
PE MinorOperatingSystemVersion	0
PE Machine	332
PE TimeDateStamp	Sat Jan 19 05:30:58 2008

Figure 48: Memory dump's basic information

b. List of running processes

Figure 49 showcases the command to acquire and print the list of running processes into a text file named “pslist.txt”. Additionally, Figure 50 showcases the contents of the file.

```
(kali@kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.pslist > pslist.txt
```

Figure 49: Command to show list of running processes

```
1|volatility 3 Framework 2.26.2
2
3 PID      PPID      ImageFileName      Offset(V)      Threads Handles SessionId      Wow64      CreateTime      ExitTime      File output
4
5 4         0         System             0x82db0910     100          541      N/A      False      2014-01-08 02:17:35.000000 UTC      N/A      Disabled
6 404       4         smss.exe           0x8454c118     4            28      N/A      False      2014-01-08 02:17:35.000000 UTC      N/A      Disabled
7 472       460       csrss.exe           0x84561968     11          466      0      False      2014-01-08 02:17:35.000000 UTC      N/A      Disabled
8 516       508       csrss.exe           0x84450770     10          305      1      False      2014-01-08 02:17:36.000000 UTC      N/A      Disabled
9 524       460       wininit.exe         0x84453770     3           98      0      False      2014-01-08 02:17:36.000000 UTC      N/A      Disabled
10 552       508       winlogon.exe        0x84465770     3          116      1      False      2014-01-08 02:17:36.000000 UTC      N/A      Disabled
11 604       524       services.exe        0x83632170     6          250      0      False      2014-01-08 02:17:36.000000 UTC      N/A      Disabled
12 616       524       lsass.exe           0x844bf770     13          610      0      False      2014-01-08 02:17:36.000000 UTC      N/A      Disabled
13 624       524       lsm.exe             0x844c2680     10          208      0      False      2014-01-08 02:17:36.000000 UTC      N/A      Disabled
14 788       604       svchost.exe         0x84866d50     6          298      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
15 848       604       svchost.exe         0x845f37a8     8          280      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
16 884       604       svchost.exe         0x848fa118     15          274      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
17 976       604       svchost.exe         0x84914d90     6          152      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
18 1000      604       svchost.exe         0x8491bd90     45         2072      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
19 1056      604       SLsvc.exe           0x8492a6d0     4           96      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
20 1088      604       svchost.exe         0x84937d90     17          567      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
21 1160      604       svchost.exe         0x84941d90     20          265      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
22 1188      604       svchost.exe         0x84945c30     22          596      0      False      2014-01-08 02:17:42.000000 UTC      N/A      Disabled
23 1308      604       svchost.exe         0x8496e9f0     17          265      0      False      2014-01-08 02:17:43.000000 UTC      N/A      Disabled
24 1424      604       spoolsv.exe         0x849c18a8     17          291      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
25 1460      604       armsvc.exe          0x849d7610     2           56      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
26 1480      604       dns.exe             0x849dcd90     10          164      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
27 1508      604       ftpbasicssvr.exe    0x849e1cc0     2           52      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
28 1576      604       svchost.exe         0x849f5888     5          124      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
29 1604      604       svchost.exe         0x849faad8     3           73      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
30 1620      604       snmp.exe            0x849f7380     4          147      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
31 1640      604       vmtoolsd.exe        0x84a0b020     7          273      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
32 1696      604       svchost.exe         0x84a21d90     4           44      0      False      2014-01-08 02:17:49.000000 UTC      N/A      Disabled
33 832       604       TPAutoConnSvc.exe   0x83066b68     9          136      0      False      2014-01-08 02:17:52.000000 UTC      N/A      Disabled
34 1924      604       dllhost.exe         0x84ba8d90     13          240      0      False      2014-01-08 02:17:53.000000 UTC      N/A      Disabled
35 1572      604       msdtc.exe           0x84bb7ca8     11          167      0      False      2014-01-08 02:17:53.000000 UTC      N/A      Disabled
36 2096      1000      taskeng.exe         0x84ba1c30     5           137      0      False      2014-01-08 02:17:53.000000 UTC      N/A      Disabled
37 2352      1000      taskeng.exe         0x84c13020     10          250      1      False      2014-01-08 02:18:17.000000 UTC      N/A      Disabled
38 2368      552       userinit.exe        0x84c148e8     0           -        1      False      2014-01-08 02:18:17.000000 UTC      2014-01-08 02:18:43.000000 UTC      Disabled
39 2392      1160      dwm.exe             0x84c1d020     3           76      1      False      2014-01-08 02:18:17.000000 UTC      N/A      Disabled
40 2480      832       TPAutoConnect.exe   0x84c2c898     2          103      1      False      2014-01-08 02:18:17.000000 UTC      N/A      Disabled
41 2496      2368      explorer.exe        0x84c2c020     24          689      1      False      2014-01-08 02:18:17.000000 UTC      N/A      Disabled
42 2580      2496      vmtoolsd.exe        0x84c5e020     6          9004      1      False      2014-01-08 02:18:18.000000 UTC      N/A      Disabled
43 2592      2496      AdobeARM.exe        0x84c5f020     6          282      1      False      2014-01-08 02:18:18.000000 UTC      N/A      Disabled
44 2616      2592      reader_sl.exe       0x84c537b0     0           -        1      False      2014-01-08 02:18:18.000000 UTC      2014-01-08 02:19:20.000000 UTC      Disabled
45 2720      2444      Oobe.exe            0x84c11298     0           -        1      False      2014-01-08 02:18:22.000000 UTC      2014-01-08 02:55:43.000000 UTC      Disabled
46 3224      604       svchost.exe         0x84c73b60     9          228      0      False      2014-01-08 02:19:53.000000 UTC      N/A      Disabled
47 3336      788       iashost.exe         0x84ce3020     2           97      0      False      2014-01-08 02:19:53.000000 UTC      N/A      Disabled
48 3680      1000      wuauclt.exe         0x84bd8020     2          139      1      False      2014-01-08 02:20:55.000000 UTC      N/A      Disabled
49 3920      2496      notepad.exe         0x84c64a50     1           51      1      False      2014-01-08 03:19:07.000000 UTC      N/A      Disabled
50 1800      2496      FTK Imager.exe      0x84cfd958     5          251      1      False      2014-01-08 03:19:32.000000 UTC      N/A      Disabled
51 1888      2496      iexplore.exe        0x848ab618     14          641      1      False      2014-01-08 03:20:24.000000 UTC      N/A      Disabled
52 2708      2496      notepad.exe         0x848a1340     1           45      1      False      2014-01-08 17:33:08.000000 UTC      N/A      Disabled
53
```

Figure 50: Memory dump's list of running processes

c. List of processes with detailed information

Figure 51 showcases the command to gather the list of processes with detailed information and print the output into a file named “psscan.txt”. Figure 52 showcase the contents of the text file.

```
(kali㉿kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.psscan > psscan.txt
```

Figure 51: Command to show processes with detailed information

```
1 Volatility 3 Framework 2.26.2
2
3 PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime File output
4
5 788 604 svchost.exe 0xc4ad50 7 298 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
6 2616 2592 reader_sl.exe 0xc817b0 0 - 1 False 2014-01-08 02:18:18.000000 UTC 2014-01-08 02:19:20.000000 UTC Disabled
7 4 0 System 0x2fb0910 100 541 N/A False 2014-01-08 02:17:35.000000 UTC N/A Disabled
8 832 604 TPAutoConnSvc.e 0x4ff9b68 9 136 0 False 2014-01-08 02:17:52.000000 UTC N/A Disabled
9 832 604 TPAutoConnSvc.e 0x7f68b68 9 136 0 False 2014-01-08 02:17:52.000000 UTC N/A Disabled
10 832 604 TPAutoConnSvc.e 0x886bb68 9 136 0 False 2014-01-08 02:17:52.000000 UTC N/A Disabled
11 2720 2444 Oobe.exe 0x1e011298 0 - 1 False 2014-01-08 02:18:22.000000 UTC 2014-01-08 02:55:43.000000 UTC Disabled
12 2352 1000 taskeng.exe 0x1e013020 10 250 1 False 2014-01-08 02:18:17.000000 UTC N/A Disabled
13 2368 552 userinit.exe 0x1e0148e8 0 - 1 False 2014-01-08 02:18:17.000000 UTC 2014-01-08 02:18:43.000000 UTC Disabled
14 2392 1160 dwm.exe 0x1e01d020 3 76 1 False 2014-01-08 02:18:17.000000 UTC N/A Disabled
15 2496 2368 explorer.exe 0x1e02c020 24 689 1 False 2014-01-08 02:18:17.000000 UTC N/A Disabled
16 2480 832 TPAutoConnect.e 0x1e02c898 2 103 1 False 2014-01-08 02:18:17.000000 UTC N/A Disabled
17 2616 2592 reader_sl.exe 0x1e0537b0 0 - 1 False 2014-01-08 02:18:18.000000 UTC 2014-01-08 02:19:20.000000 UTC Disabled
18 2580 2496 vmtoolsd.exe 0x1e05e020 6 9004 1 False 2014-01-08 02:18:18.000000 UTC N/A Disabled
19 2592 2496 AdobeARM.exe 0x1e05f020 6 282 1 False 2014-01-08 02:18:18.000000 UTC N/A Disabled
20 3920 2496 notepad.exe 0x1e064a50 1 51 1 False 2014-01-08 03:19:07.000000 UTC N/A Disabled
21 3224 604 svchost.exe 0x1e073b60 9 228 0 False 2014-01-08 02:19:53.000000 UTC N/A Disabled
22 3336 788 iashost.exe 0x1e0e3020 2 97 0 False 2014-01-08 02:19:53.000000 UTC N/A Disabled
23 1800 2496 FTK Imager.exe 0x1e0fd958 5 251 1 False 2014-01-08 03:19:32.000000 UTC N/A Disabled
24 1640 604 vmtoolsd.exe 0x1e20b020 7 273 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
25 1696 604 svchost.exe 0x1e221d00 4 44 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
26 2096 1000 taskeng.exe 0x1e3a1c30 5 137 0 False 2014-01-08 02:17:53.000000 UTC N/A Disabled
27 1924 604 dllhost.exe 0x1e3a8d90 13 240 0 False 2014-01-08 02:17:53.000000 UTC N/A Disabled
28 1572 604 msdtc.exe 0x1e3b7ca8 11 167 0 False 2014-01-08 02:17:53.000000 UTC N/A Disabled
29 3680 1000 wuauclt.exe 0x1e3d8020 2 139 1 False 2014-01-08 02:20:55.000000 UTC N/A Disabled
30 788 604 svchost.exe 0x1e466d50 6 298 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
31 2708 2496 notepad.exe 0x1e4a1340 1 45 1 False 2014-01-08 17:33:08.000000 UTC N/A Disabled
32 1888 2496 iexplore.exe 0x1e4ab618 14 641 1 False 2014-01-08 03:20:24.000000 UTC N/A Disabled
33 884 604 svchost.exe 0x1e4fa118 15 274 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
34 976 604 svchost.exe 0x1e514d90 6 152 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
35 1000 604 svchost.exe 0x1e51bd90 45 2072 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
36 1056 604 SLsvc.exe 0x1e52a6d0 4 96 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
37 1088 604 svchost.exe 0x1e537d90 17 567 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
38 1160 604 svchost.exe 0x1e541d90 20 265 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
39 1188 604 svchost.exe 0x1e545c30 22 596 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
40 1308 604 svchost.exe 0x1e56e9f0 17 265 0 False 2014-01-08 02:17:43.000000 UTC N/A Disabled
41 1424 604 spoolsv.exe 0x1e5c18a8 17 291 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
42 1460 604 armsvc.exe 0x1e5d7610 2 56 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
43 1480 604 dns.exe 0x1e5dcd90 10 164 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
44 1508 604 ftpbasicsvr.exe 0x1e5e1cc0 2 52 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
45 1576 604 svchost.exe 0x1e5f5888 5 124 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
46 1620 604 snmp.exe 0x1e5f7380 4 147 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
47 1604 604 svchost.exe 0x1e5faad8 3 73 0 False 2014-01-08 02:17:49.000000 UTC N/A Disabled
48 516 508 csrss.exe 0x1e850770 10 305 1 False 2014-01-08 02:17:36.000000 UTC N/A Disabled
49 524 460 wininit.exe 0x1e853770 3 98 0 False 2014-01-08 02:17:36.000000 UTC N/A Disabled
50 552 508 winlogon.exe 0x1e865770 3 116 1 False 2014-01-08 02:17:36.000000 UTC N/A Disabled
51 616 524 lsass.exe 0x1e8bf770 13 610 0 False 2014-01-08 02:17:36.000000 UTC N/A Disabled
52 624 524 lsm.exe 0x1e8c2680 10 208 0 False 2014-01-08 02:17:36.000000 UTC N/A Disabled
53 404 4 smss.exe 0x1e94c118 4 28 N/A False 2014-01-08 02:17:35.000000 UTC N/A Disabled
54 472 460 csrss.exe 0x1e961968 11 466 0 False 2014-01-08 02:17:35.000000 UTC N/A Disabled
55 848 604 svchost.exe 0x1e9f37a8 8 280 0 False 2014-01-08 02:17:42.000000 UTC N/A Disabled
56 604 524 services.exe 0x1f632170 6 250 0 False 2014-01-08 02:17:36.000000 UTC N/A Disabled
57 832 604 TPAutoConnSvc.e 0x1fc66b68 9 136 0 False 2014-01-08 02:17:52.000000 UTC N/A Disabled
58
```

Figure 52: List of processes with detailed information

Figure 53 showcase the command to get the list of handles of processes and write them to a text file named “handles.txt”. Figure 54 showcases the contents of the file.

```
(kali㉿kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.handles > handles.txt
```

Figure 53: Command to show list of handles

PID	Process	Offset	HandleValue	Type	GrantedAccess	Name
5	System	0x82db0910	0x4	Process	0x1fffff	System Pid 4
6	System	0x86213398	0x8	Key	0x2001f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\HIVELIST
7	System	0x8620aa90	0xc	Key	0x0	-
8	System	0x8625e770	0x10	Key	0x2001f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\PRODUCTOPTIONS
9	System	0x8625cd68	0x14	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\SESSION MANAGER\MEMORY MANAGEMENT_PREFETCHPARAMETERS
10	System	0x87a4ada0	0x18	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
11	System	0x86261188	0x1c	Key	0x2001f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
12	System	0x862d7138	0x20	Key	0x20019	MACHINE\SYSTEM\CONTROLSET001\CONTROL\WMI\SECURITY
13	System	0x86246210	0x24	Key	0x20019	MACHINE\SYSTEM\CONTROLSET001\CONTROL\WMI\SECURITY
14	System	0x82ee33b0	0x28	Thread	0x1fffff	Tid 184 Pid 4
15	System	0x83ee14e0	0x2c	Thread	0x0	Tid 304 Pid 4
16	System	0x88406f18	0x30	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\INITVERBS
17	System	0x86368902	0x34	Token	0xe	-
18	System	0x8842d938	0x38	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
19	System	0x863ba830	0x3c	Key	0x20019	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
20	System	0x88419690	0x40	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
21	System	0x862d5020	0x44	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
22	System	0x884011d0	0x48	Key	0x20019	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
23	System	0x875a2480	0x4c	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
24	System	0x8625d3a8	0x50	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
25	System	0x88423eb0	0x54	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
26	System	0x8633a850	0x58	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
27	System	0x836b1b40	0x5c	File	0x120116	\Device\Npfs
28	System	0x8841c6f0	0x60	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
29	System	0x8362da08	0x64	TmRm	0x12007f	-
30	System	0x8362a810	0x68	TmTm	0xf003f	-
31	System	0x8362d028	0x6c	TmRm	0x1f007f	-
32	System	0x8362a6b0	0x70	File	0x12019f	\Device\NpfsKtmLog
33	System	0x83629b78	0x74	File	0x13019f	\Device\Npfs\Device\HarddiskVolume1\Extend\%\$RmMetadata%\\$TxFLog\%\$TxFLog
34	System	0x83623d90	0x78	File	0x12019f	\Device\Npfs\Device\HarddiskVolume1\Extend\%\$RmMetadata%\\$TxFLog\%\$TxFLog
35	System	0x835e0408	0x7c	File	0x1	\Device\HarddiskVolume1
36	System	0x8314a600	0x80	File	0x13019f	\Device\Npfs\Device\HarddiskVolume1\Extend\%\$RmMetadata%\\$TxFLog\%\$TxFLog
37	System	0x835e0870	0x84	File	0x12019f	\Device\HarddiskVolume1\Extend\%\$RmMetadata%\\$TxFLog\%\$TxFLogContainer00000000000000000000
38	System	0x835e0918	0x88	File	0x12019f	\Device\HarddiskVolume1\Extend\%\$RmMetadata%\\$TxFLog\%\$TxFLogContainer00000000000000000000
39	System	0x831672e0	0x8c	File	0x12019f	\Device\HarddiskVolume1\Extend\%\$RmMetadata%\\$TxFLog\%\$TxFLog.blf
40	System	0x83629530	0x90	Thread	0x1fffff	Tid 232 Pid 4
41	System	0x8372aa68	0x94	File	0x12019f	\Device\NpfsTxFLog
42	System	0x8625a350	0x98	Key	0xf003f	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
43	System	0x86202670	0x9c	Token	0xf01ff	-
44	System	0x82db0910	0xa0	Process	0x1440	System Pid 4
45	System	0x863b5518	0xa4	Key	0x20019	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
46	System	0x8633a8a0	0xa8	Key	0x10	MACHINE\SYSTEM\CONTROLSET001\CONTROL\NETWORKPROVIDER\ORDER
47	System	0x862bfa78	0xac	Key	0x20019	MACHINE\SYSTEM\CONTROLSET001\CONTROL\CLASS\{4d36e96c-e325-11ce-bfc1-08002be10318}\0000\MIXERSETTINGS
48	System	0x863f57d8				

Figure 54: List of handles

e. Handles of the process with PID=1056

Figure 55 showcases the command to show the list of handles of the process with PID 1056, along with the output of the command.

```

L$ python3 vol.py -f a1memorydump.mem windows.handles --pid 1056
Volatility 3 Framework 2.26.2
Progress: 100.00 PDB scanning finished
PID Process Offset HandleValue Type GrantedAccess Name
1056 SLsvc.exe 0x89cd9948 0x4 Directory 0x3 KnownDlls
1056 SLsvc.exe 0x8492c340 0x8 File 0x100020 \Device\HarddiskVolume1\Windows\System
32
1056 SLsvc.exe 0x8492a438 0xc Event 0x1f0003 -
1056 SLsvc.exe 0x8492a3f8 0x10 Mutant 0x1f0001 -
1056 SLsvc.exe 0x8492b4a8 0x14 ALPC Port 0x1f0001 -
1056 SLsvc.exe 0x863ad318 0x18 Key 0x20019 MACHINE
1056 SLsvc.exe 0x8492d098 0x1c Event 0x1f0003 -
1056 SLsvc.exe 0x862d06e0 0x20 Key 0x1 MACHINE\SYSTEM\CONTROLSET001\CONTROL\SESSION M
ANAGER
1056 SLsvc.exe 0x845f69f0 0x24 WindowStation 0xf006e Service-0x0-3e4$
1056 SLsvc.exe 0x8492d310 0x28 Event 0x21f0003 -
1056 SLsvc.exe 0x845f6930 0x2c Desktop 0xf00cf Default
1056 SLsvc.exe 0x845f69f0 0x30 WindowStation 0xf006e Service-0x0-3e4$
1056 SLsvc.exe 0x8493310c 0x34 EtwRegistration 0x804 -
1056 SLsvc.exe 0x8492da08 0x38 Event 0x1f0003 -
1056 SLsvc.exe 0x849331068 0x3c EtwRegistration 0x804 -
1056 SLsvc.exe 0x80ca3cc0 0x40 Token 0x48 -
1056 SLsvc.exe 0x84932e98 0x44 File 0x12019f \Device\NamedPipe\nt\NtControlPipe6
1056 SLsvc.exe 0x89d99330 0x48 Directory 0xf BaseNamedObjects
1056 SLsvc.exe 0x84932d80 0x4c Event 0x1f0003 -
1056 SLsvc.exe 0x84932d50 0x50 Event 0x1f0003 -
1056 SLsvc.exe 0x84932d20 0x54 Event 0x1f0003 -
1056 SLsvc.exe 0x84932cd0 0x58 EtwRegistration 0x804 -
1056 SLsvc.exe 0x84932c98 0x5c Event 0x1f0003 -
1056 SLsvc.exe 0x8492c030 0x60 Thread 0x1fffff Tid 1060 Pid 1056
1056 SLsvc.exe 0x84932b80 0x64 ALPC Port 0x1f0001 -
1056 SLsvc.exe 0x849338c8 0x70 Event 0x1f0003 -
1056 SLsvc.exe 0x84934728 0x74 Event 0x1f0003 -
1056 SLsvc.exe 0x849328a8 0x78 Event 0x1f0003 -
1056 SLsvc.exe 0x8492e758 0x7c Timer 0x1f0003 -
1056 SLsvc.exe 0x84933620 0x80 Thread 0x1fffff Tid 1084 Pid 1056
1056 SLsvc.exe 0x84933620 0x84 Thread 0x1fffff Tid 1084 Pid 1056
1056 SLsvc.exe 0x913be800 0x88 KeyedEvent 0xf0003 -
1056 SLsvc.exe 0x84934020 0x8c IoCompletion 0x1f0003 -
1056 SLsvc.exe 0x84934510 0x90 TpWorkerFactory 0xf00ff -
1056 SLsvc.exe 0x88431158 0x94 Section 0xf0007 DSEC420
1056 SLsvc.exe 0x84937bd0 0x98 ALPC Port 0x1f0001 SLCTransportEndpoint-00001
1056 SLsvc.exe 0x84937b70 0x9c IoCompletion 0x1f0003 -
1056 SLsvc.exe 0x84937b20 0xa0 IoCompletion 0x1f0003 -
1056 SLsvc.exe 0x84937b70 0xa4 IoCompletion 0x1f0003 -
1056 SLsvc.exe 0x845e93e0 0xa8 Event 0x1f0003 -
1056 SLsvc.exe 0x84939ac0 0xac Thread 0x1fffff Tid 1100 Pid 1056
1056 SLsvc.exe 0x8493b020 0xb0 ALPC Port 0x1f0001 -
1056 SLsvc.exe 0x9138e078 0xb4 KeyedEvent 0xf0003 -
1056 SLsvc.exe 0x8491a020 0xb8 Timer 0x100002 -
1056 SLsvc.exe 0x84938228 0xbc Timer 0x100002 -
1056 SLsvc.exe 0x8495c870 0xc0 EtwRegistration 0x804 -

```

Figure 55: Handles of process with PID=1056

f. List of hidden processes

Figure 56 showcases the command to acquire the list of hidden processes and write the output into a file named “hidden.txt”. Figure 57 shows the contents of the file.

```
(kali㉿kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.psxview > hidden.txt
```

Figure 56: Command to show list of hidden processes

```
1 Volatility 3 Framework 2.26.2
2
3 Offset(Virtual) Name PID pslist psscan thrdsan csrss Exit Time
4
5 0x849d7610 armsvc.exe 1460 True False True True
6 0x1e5d7610 armsvc.exe 1460 False True False False
7 0x848ab618 iexplore.exe 1888 True False True True
8 0x1e4ab618 iexplore.exe 1888 False True False False
9 0x1e3d8020 wuauclt.exe 3680 False True False False
10 0x1e01d020 dwm.exe 2392 False True False False
11 0x84a0b020 vmtoolsd.exe 1640 True False True True
12 0x84c13020 taskeng.exe 2352 True False True True
13 0x84c1d020 dwm.exe 2392 True False True True
14 0x84c2c020 explorer.exe 2496 True False True True
15 0x84c5e020 vmtoolsd.exe 2580 True False True True
16 0x84c5f020 AdobeARM.exe 2592 True False True True
17 0x84ce3020 iashost.exe 3336 True False True True
18 0x84bd8020 wuauclt.exe 3680 True False True True
19 0x1e013020 taskeng.exe 2352 False True False False
20 0x1e05f020 AdobeARM.exe 2592 False True False False
21 0x84945c30 svchost.exe 1188 True False True True
22 0x84ba1c30 taskeng.exe 2096 True False True True
23 0x1e3a1c30 taskeng.exe 2096 False True False False
24 0x1e545c30 svchost.exe 1188 False True False False
25 0x1e0e3020 iashost.exe 3336 False True False False
26 0x1e5dcd90 dns.exe 1480 False True False False
27 0x84c64a50 notepad.exe 3920 True False True True
28 0x1e064a50 notepad.exe 3920 False True False False
29 0x844c2680 lsm.exe 624 True False True True
30 0x1e8c2680 lsm.exe 624 False True False False
31 0x849f5888 svchost.exe 1576 True False True True
32 0x1e5f5888 svchost.exe 1576 False True False False
33 0x84c2c898 TPAutoConnect.e 2480 True False True True
34 0x84c11298 Oobe.exe 2720 True False False False 2014-01-08 02:55:43+00:00
35 0x1e011298 Oobe.exe 2720 False True False False 2014-01-08 02:55:43+00:00
36 0x1e02c898 TPAutoConnect.e 2480 False True False False
37 0x1e02c020 explorer.exe 2496 False True False False
38 0x849c18a8 spoolsv.exe 1424 True False True True
39 0x84bb7ca8 msdtc.exe 1572 True False True True
40 0x1e3b7ca8 msdtc.exe 1572 False True False False
41 0x1e5c18a8 spoolsv.exe 1424 False True False False
42 0x1e541d90 svchost.exe 1160 False True False False
43 0x849e1cc0 ftpbasicsvr.exe 1508 True False True True
44 0x1e5e1cc0 ftpbasicsvr.exe 1508 False True False False
45 0x8492a6d0 SLsvc.exe 1056 True False True True
```

Figure 57: List of hidden processes

g. Registry hives

Figure 58 shows the command to acquire the registry hives, along with the output of the command.


```
(kali@kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.registry.hivelist
Volatility 3 Framework 2.26.2
Progress: 100.00 PDB scanning finished
Offset FileFullPath File output
0x812eb6b0 \Device\HarddiskVolume1\Windows\ServiceProfiles\NetworkService\NTUSER.DAT Disabled
0x81321008 \Device\HarddiskVolume1\Windows\ServiceProfiles\LocalService\NTUSER.DAT Disabled
0x86211008 Disabled
0x86226008 \REGISTRY\MACHINE\SYSTEM Disabled
0x86248008 \REGISTRY\MACHINE\HARDWARE Disabled
0x89c2f148 \Device\HarddiskVolume1\Windows\System32\config\DEFAULT Disabled
0x89c33450 \Device\HarddiskVolume1\Windows\System32\config\SAM Disabled
0x89c36008 \Device\HarddiskVolume1\Windows\System32\config\SECURITY Disabled
0x89c47008 \Device\HarddiskVolume1\Windows\System32\config\COMPONENTS Disabled
0x89c47a20 \Device\HarddiskVolume1\Windows\System32\config\SOFTWARE Disabled
0x89cd1a20 \Device\HarddiskVolume1\Boot\BCD Disabled
0x9465f6a8 \Device\HarddiskVolume1\Users\Administrator\NTUSER.DAT Disabled
0x946ae008 \Device\HarddiskVolume1\Users\Administrator\AppData\Local\Microsoft\Windows\UsrClass.dat Disabled
```

Figure 58: Registry hives

h. List of Dynamic Link Libraries (DLLs)

Figure 59 shows the command to acquire the list of DLLs and write the list to a text file named “dlllist.txt”. Figure 60 showcases the contents of the file.

```
(kali@kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.dlllist > dlllist.txt
```

Figure 59: Command to show DLLs

Volatility 3 Framework 2.26.2										
3	PID	Process	Base	Size	Name	Path	LoadTime	File output		
5	404	smss.exe	0x47610000	0x12000	-	-	-	N/A	Disabled	
6	404	smss.exe	0x77d00000	0x127000	-	-	-	N/A	Disabled	
7	472	csrss.exe	0x4a130000	0x5000	csrss.exe	C:\Windows\system32\csrss.exe	N/A	Disabled		
8	472	csrss.exe	0x77d00000	0x127000	-	-	-	N/A	Disabled	
9	472	csrss.exe	0x764d0000	0xf000	CSRSRV.dll	C:\Windows\system32\CSRSRV.dll	N/A	Disabled		
10	472	csrss.exe	0x764b0000	0x13000	basesrv.dll	C:\Windows\system32\basesrv.dll	N/A	Disabled		
11	472	csrss.exe	0x76450000	0x60000	winsrv.dll	C:\Windows\system32\winsrv.dll	N/A	Disabled		
12	472	csrss.exe	0x765a0000	0x9d000	USER32.dll	C:\Windows\system32\USER32.dll	N/A	Disabled		
13	472	csrss.exe	0x76bb0000	0xdb000	KERNEL32.dll	C:\Windows\system32\KERNEL32.dll	N/A	Disabled		
14	472	csrss.exe	0x77f10000	0x4b000	GDI32.dll	C:\Windows\system32\GDI32.dll	N/A	Disabled		
15	472	csrss.exe	0x779d0000	0xc6000	ADVAPI32.dll	C:\Windows\system32\ADVAPI32.dll	N/A	Disabled		
16	472	csrss.exe	0x77c20000	0xc3000	RPCRT4.dll	C:\Windows\system32\RPCRT4.dll	N/A	Disabled		
17	472	csrss.exe	0x76580000	0x9000	LPK.DLL	C:\Windows\system32\LPK.DLL	N/A	Disabled		
18	472	csrss.exe	0x77b50000	0x7d000	USP10.dll	C:\Windows\system32\USP10.dll	N/A	Disabled		
19	472	csrss.exe	0x77e60000	0xaa000	msvcrt.dll	C:\Windows\system32\msvcrt.dll	N/A	Disabled		
20	472	csrss.exe	0x762c0000	0x5f000	sxs.dll	C:\Windows\system32\sxs.dll	N/A	Disabled		
21	516	csrss.exe	0x4a130000	0x5000	csrss.exe	C:\Windows\system32\csrss.exe	N/A	Disabled		
22	516	csrss.exe	0x77d00000	0x127000	-	-	-	N/A	Disabled	
23	516	csrss.exe	0x764d0000	0xf000	CSRSRV.dll	C:\Windows\system32\CSRSRV.dll	N/A	Disabled		
24	516	csrss.exe	0x764b0000	0x13000	basesrv.dll	C:\Windows\system32\basesrv.dll	N/A	Disabled		
25	516	csrss.exe	0x76450000	0x60000	winsrv.dll	C:\Windows\system32\winsrv.dll	N/A	Disabled		
26	516	csrss.exe	0x765a0000	0x9d000	USER32.dll	C:\Windows\system32\USER32.dll	N/A	Disabled		
27	516	csrss.exe	0x76bb0000	0xdb000	KERNEL32.dll	C:\Windows\system32\KERNEL32.dll	N/A	Disabled		
28	516	csrss.exe	0x77f10000	0x4b000	GDI32.dll	C:\Windows\system32\GDI32.dll	N/A	Disabled		
29	516	csrss.exe	0x779d0000	0xc6000	ADVAPI32.dll	C:\Windows\system32\ADVAPI32.dll	N/A	Disabled		
30	516	csrss.exe	0x77c20000	0xc3000	RPCRT4.dll	C:\Windows\system32\RPCRT4.dll	N/A	Disabled		
31	516	csrss.exe	0x76580000	0x9000	LPK.DLL	C:\Windows\system32\LPK.DLL	N/A	Disabled		
32	516	csrss.exe	0x77b50000	0x7d000	USP10.dll	C:\Windows\system32\USP10.dll	N/A	Disabled		
33	516	csrss.exe	0x77e60000	0xaa000	msvcrt.dll	C:\Windows\system32\msvcrt.dll	N/A	Disabled		
34	516	csrss.exe	0x762c0000	0x5f000	sxs.dll	C:\Windows\system32\sxs.dll	N/A	Disabled		
35	604	services.exe	0x710000	0x47000	services.exe	C:\Windows\system32\services.exe	N/A	Disabled		
36	604	services.exe	0x77d00000	0x127000	ntdll.dll	C:\Windows\system32\ntdll.dll	N/A	Disabled		
37	604	services.exe	0x76bb0000	0xdb000	kernel32.dll	C:\Windows\system32\kernel32.dll	N/A	Disabled		
38	604	services.exe	0x779d0000	0xc6000	ADVAPI32.dll	C:\Windows\system32\ADVAPI32.dll	N/A	Disabled		
39	604	services.exe	0x77c20000	0xc3000	RPCRT4.dll	C:\Windows\system32\RPCRT4.dll	N/A	Disabled		
40	604	services.exe	0x765a0000	0x9d000	USER32.dll	C:\Windows\system32\USER32.dll	N/A	Disabled		
41	604	services.exe	0x77f10000	0x4b000	GDI32.dll	C:\Windows\system32\GDI32.dll	N/A	Disabled		
42	604	services.exe	0x77e60000	0xaa000	msvcrt.dll	C:\Windows\system32\msvcrt.dll	N/A	Disabled		
43	604	services.exe	0x76410000	0x1e000	USERENV.dll	C:\Windows\system32\USERENV.dll	N/A	Disabled		
44	604	services.exe	0x76430000	0x14000	Secur32.dll	C:\Windows\system32\Secur32.dll	N/A	Disabled		
45	604	services.exe	0x76330000	0x4e000	SCESRV.dll	C:\Windows\system32\SCESRV.dll	N/A	Disabled		

Figure 60: List of DLLs

i. List of DLLs that are being used by PID 2708

Figure 61 showcases the command to get the list of DLLs being used by the process with PID 2708 and write the output of the command into a csv file named “dll_pid2708”.

Figure 62 showcases the contents of the file on the terminal via the command “cat dll_pid2708.csv”.

```
(kali㉿kali)-[~/volatility3]
$ python3 vol.py -f a1memorydump.mem windows.dllexport --pid 2708 > dll_pid2708.csv
```

Figure 61: Command to show list of DLLs being used by PID=2708

```
1|Volatility 3 Framework 2.26.2
2
3 PID      Process Base      Size      Name      Path      LoadTime      File output
4
5 2708     notepad.exe      0xb90000      0x28000 notepad.exe C:\Windows\System32\notepad.exe N/A      Disabled
6 2708     notepad.exe      0x77d00000      0x127000 ntdll.dll C:\Windows\system32\ntdll.dll N/A      Disabled
7 2708     notepad.exe      0x76bb0000      0xdb000 kernel32.dll C:\Windows\system32\kernel32.dll N/A      Disabled
8 2708     notepad.exe      0x779d0000      0xc6000 ADVAPI32.dll C:\Windows\system32\ADVAPI32.dll N/A      Disabled
9 2708     notepad.exe      0x77c20000      0xc3000 RPCRT4.dll C:\Windows\system32\RPCRT4.dll N/A      Disabled
10 2708     notepad.exe      0x77f10000      0x4b000 GDI32.dll C:\Windows\system32\GDI32.dll N/A      Disabled
11 2708     notepad.exe      0x765a0000      0x9d000 USER32.dll C:\Windows\system32\USER32.dll N/A      Disabled
12 2708     notepad.exe      0x77e60000      0xaa000 msvcrt.dll C:\Windows\system32\msvcrt.dll N/A      Disabled
13 2708     notepad.exe      0x76870000      0x73000 COMDLG32.dll C:\Windows\system32\COMDLG32.dll N/A      Disabled
14 2708     notepad.exe      0x76cc0000      0x58000 SHLWAPI.dll C:\Windows\system32\SHLWAPI.dll N/A      Disabled
15 2708     notepad.exe      0x75150000      0x19e000 COMCTL32.dll C:\Windows\WinSxS\x86_microsoft.windows.common-controls_65
16 2708     notepad.exe      0x76e70000      0xb0f000 SHELL32.dll C:\Windows\system32\SHELL32.dll N/A      Disabled
17 2708     notepad.exe      0x73780000      0x42000 WINSPOOL.DRV C:\Windows\System32\WINSPOOL.DRV N/A      Disabled
18 2708     notepad.exe      0x76d20000      0x144000 ole32.dll C:\Windows\system32\ole32.dll N/A      Disabled
19 2708     notepad.exe      0x767e0000      0x8d000 OLEAUT32.dll C:\Windows\system32\OLEAUT32.dll N/A      Disabled
20 2708     notepad.exe      0x77b30000      0x1e000 IMM32.DLL C:\Windows\system32\IMM32.DLL N/A      Disabled
21 2708     notepad.exe      0x76640000      0xc8000 MSCTF.dll C:\Windows\system32\MSCTF.dll N/A      Disabled
22 2708     notepad.exe      0x76580000      0x9000 LPK.DLL C:\Windows\system32\LPK.DLL N/A      Disabled
23 2708     notepad.exe      0x77b50000      0x7d000 USP10.dll C:\Windows\system32\USP10.dll N/A      Disabled
24 2708     notepad.exe      0x756a0000      0x3f000 UxTheme.dll C:\Windows\System32\UxTheme.dll N/A      Disabled
25
```

Figure 62: List of DLLs being used by PID=2708 - shown in csv

```
(kali㉿kali)-[~/volatility3]
$ cat dll_pid2708.csv
Volatility 3 Framework 2.26.2
```

PID	Process	Base	Size	Name	Path	LoadTime	File	output
2708	notepad.exe	0xb90000	0x28000	notepad.exe	C:\Windows\System32\notepad.exe	N/A	Disabled	
2708	notepad.exe	0x77d00000	0x127000	ntdll.dll	C:\Windows\system32\ntdll.dll	N/A	Disabled	
2708	notepad.exe	0x76bb0000	0xdb000	kernel32.dll	C:\Windows\system32\kernel32.dll	N/A	Disabled	
2708	notepad.exe	0x779d0000	0xc6000	ADVAPI32.dll	C:\Windows\system32\ADVAPI32.dll	N/A	Disabled	
2708	notepad.exe	0x77c20000	0xc3000	RPCRT4.dll	C:\Windows\system32\RPCRT4.dll	N/A	Disabled	
2708	notepad.exe	0x77f10000	0x4b000	GDI32.dll	C:\Windows\system32\GDI32.dll	N/A	Disabled	
2708	notepad.exe	0x765a0000	0x9d000	USER32.dll	C:\Windows\system32\USER32.dll	N/A	Disabled	
2708	notepad.exe	0x77e60000	0xaa000	msvcrt.dll	C:\Windows\system32\msvcrt.dll	N/A	Disabled	
2708	notepad.exe	0x76870000	0x73000	COMDLG32.dll	C:\Windows\system32\COMDLG32.dll	N/A	Disabled	
2708	notepad.exe	0x76cc0000	0x58000	SHLWAPI.dll	C:\Windows\system32\SHLWAPI.dll	N/A	Disabled	
2708	notepad.exe	0x75150000	0x19e000	COMCTL32.dll	C:\Windows\WinSxS\x86_microsoft.windows.common-con			
2708	notepad.exe	0x76e70000	0xb0f000	SHELL32.dll	C:\Windows\system32\SHELL32.dll	N/A	Disabled	
2708	notepad.exe	0x73780000	0x42000	WINSPOOL.DRV	C:\Windows\System32\WINSPOOL.DRV	N/A	Disabled	
2708	notepad.exe	0x76d20000	0x144000	ole32.dll	C:\Windows\system32\ole32.dll	N/A	Disabled	
2708	notepad.exe	0x767e0000	0x8d000	OLEAUT32.dll	C:\Windows\system32\OLEAUT32.dll	N/A	Disabled	
2708	notepad.exe	0x77b30000	0x1e000	IMM32.DLL	C:\Windows\system32\IMM32.DLL	N/A	Disabled	
2708	notepad.exe	0x76640000	0xc8000	MSCTF.dll	C:\Windows\system32\MSCTF.dll	N/A	Disabled	
2708	notepad.exe	0x76580000	0x9000	LPK.DLL	C:\Windows\system32\LPK.DLL	N/A	Disabled	
2708	notepad.exe	0x77b50000	0x7d000	USP10.dll	C:\Windows\system32\USP10.dll	N/A	Disabled	
2708	notepad.exe	0x756a0000	0x3f000	UxTheme.dll	C:\Windows\System32\UxTheme.dll	N/A	Disabled	

Figure 63: List of DLLs being used by PID=2708 - shown in terminal